
بسم الله الرحمن الرحيم
Bismillahirrahmanirrahim


Woodcraft
(Web Application)
Virtual University

GROUP MEMBER INTRODUCTION

Submitted By:

Student Name:	**NOOR UL NISA**
VU ID:	**B**********4**
Program:	**BSIT (8th Semester)**
Project Group ID:	**F**********C**
Project Title:	**Woodcraft (Web App)**

Supervisor:

Abdul Majid Khokhar

Solo Project Developer



NOOR UL NISA

An 8th-semester BSIT student at the Virtual University of Pakistan, I have spearheaded the design and development of the Woodcraft Management System as a standalone project. My role spans full-stack engineering, database architecture, and UI/UX design.

"Combining traditional artisan craftsmanship with modern digital efficiency."

What is Woodcraft ?

Woodcraft is a professional Web-Application designed to digitize the end-to-end operations of a woodcraft workshop.

It bridges the gap between traditional craftsmanship and modern management by integrating order processing, real-time inventory, and artisan worklogs into one seamless interface.

"Empowering workshops with data-driven decision making."



1. The Digital Need in Woodworking

- **Manual Inefficiency:** Traditional workshops rely on physical registers, leading to lost data, inaccurate inventory levels, and payment discrepancies.
- **Lack of Transparency:** Customers often have no way to track progress without visiting the shop, and employees struggle to prioritize tasks without clear assignments.
- **Resource Wastage:** Without material tracking, wood and hardware are often over-purchased or unavailable when needed.

Project Objective

To build a role-aware dashboard that bridges the gap between traditional craftsmanship and modern digital efficiency through React 18 and MySQL.



Digital Transformation | Efficiency

2. Woodcraft Scope & Functional Core

- **Role-Based Ecosystem:** Personalized interfaces for Admins (Management), Customers (Orders), and Employees (Tasks).
- **Automated Workflow:** Real-time conversion of orders into digital invoices with automated balance and wage calculations.
- **Inventory Management:** Dynamic material tracking with "Low Stock" alerts to prevent workshop downtime.
- **Security & Approval:** Secure registration flow where Admins must approve new accounts before system access.



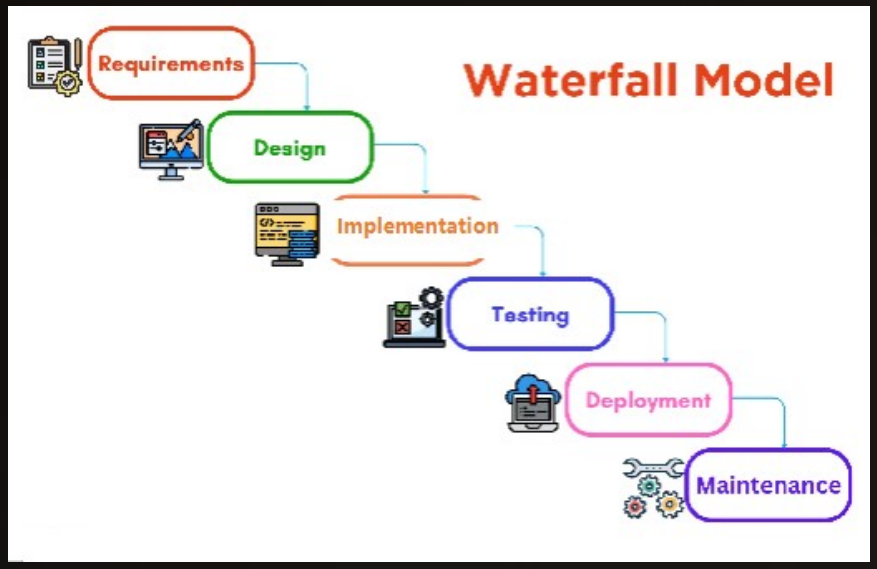
Role-Based Access Automation

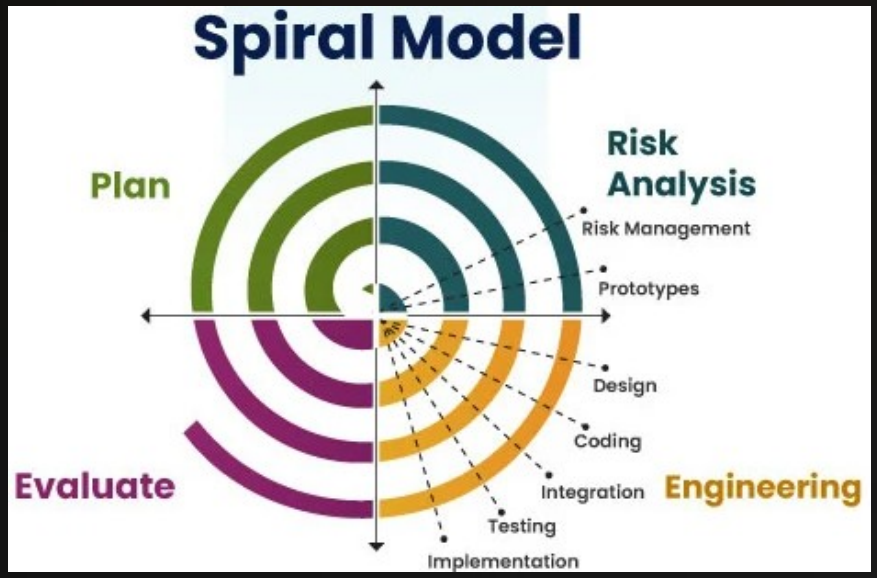
VU Process Model

*"**Woodcraft**" project follows the **VU Process Model** is a hybrid methodology combining:*

1. ***Waterfall**'s structured, phase-by-phase documentation.*
2. ***Spiral**'s iterative development and risk analysis.*

It follows phases sequentially (like Waterfall) but each phase contains mini-iterations (like Spiral).





Adopted Methodology: VU Process Model

Gathering

Requirement gathering & analysis.

Design

Architecture, Class, & DB design.

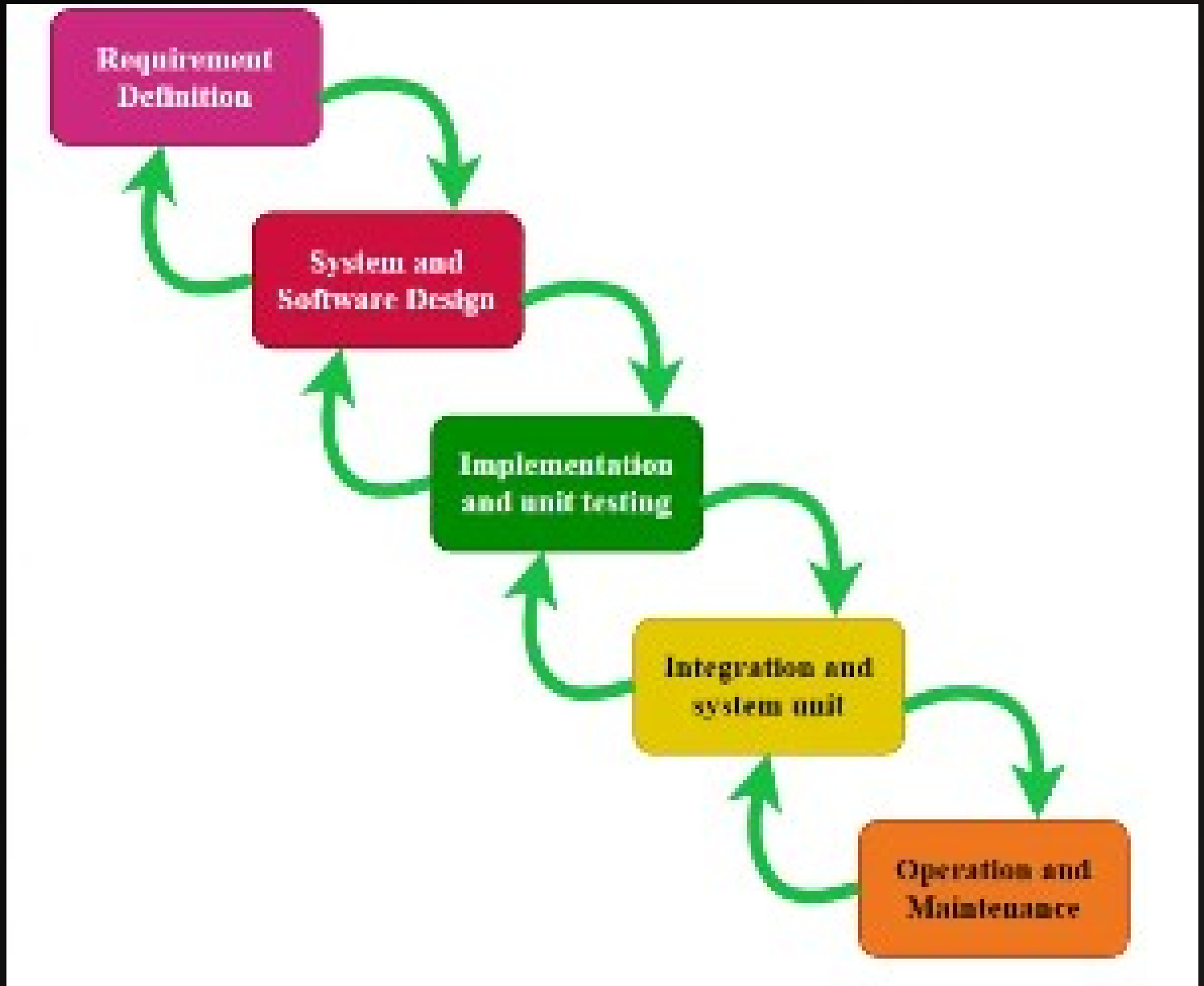
Implementation

Iterative coding and unit testing.

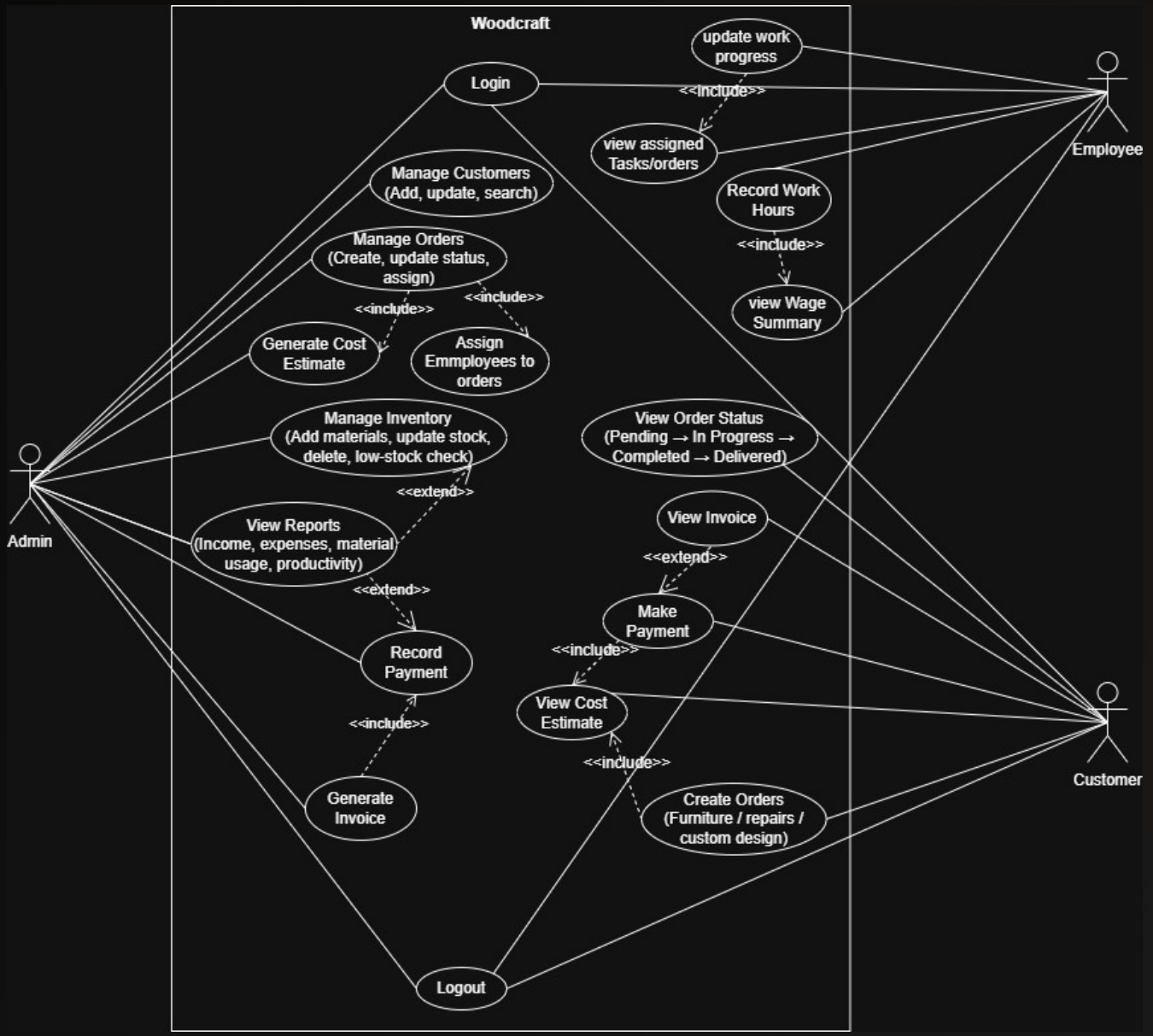
Reporting

Final documentation & evaluation.

VU Process Model



Use Case Diagram



Functional Requirements

Functional requirements define what the system must do

Core Functional Requirements

- **Order Lifecycle:** Track from Pending to Delivery status.
- **Inventory Monitoring:** Automatic stock reduction and low-stock alerts.
- **Cost Estimation:** Dynamic calculation based on materials and labor.
- **Billing:** Auto-generated invoices and payment history.



Customer & Order Lifecycle

- **Identity & Profiles:** Secure registration, authentication, and full administrative oversight of user accounts.
- **Commission Creation:** Integrated workflows for furniture orders, repair requests, and custom designs.
- **Real-time Tracking:** Live order status flow (Pending, In Progress, Delivered) with unique Order ID auditing.
- **Intelligent Estimation:** Automated cost generation based on live material stock and labor hours.

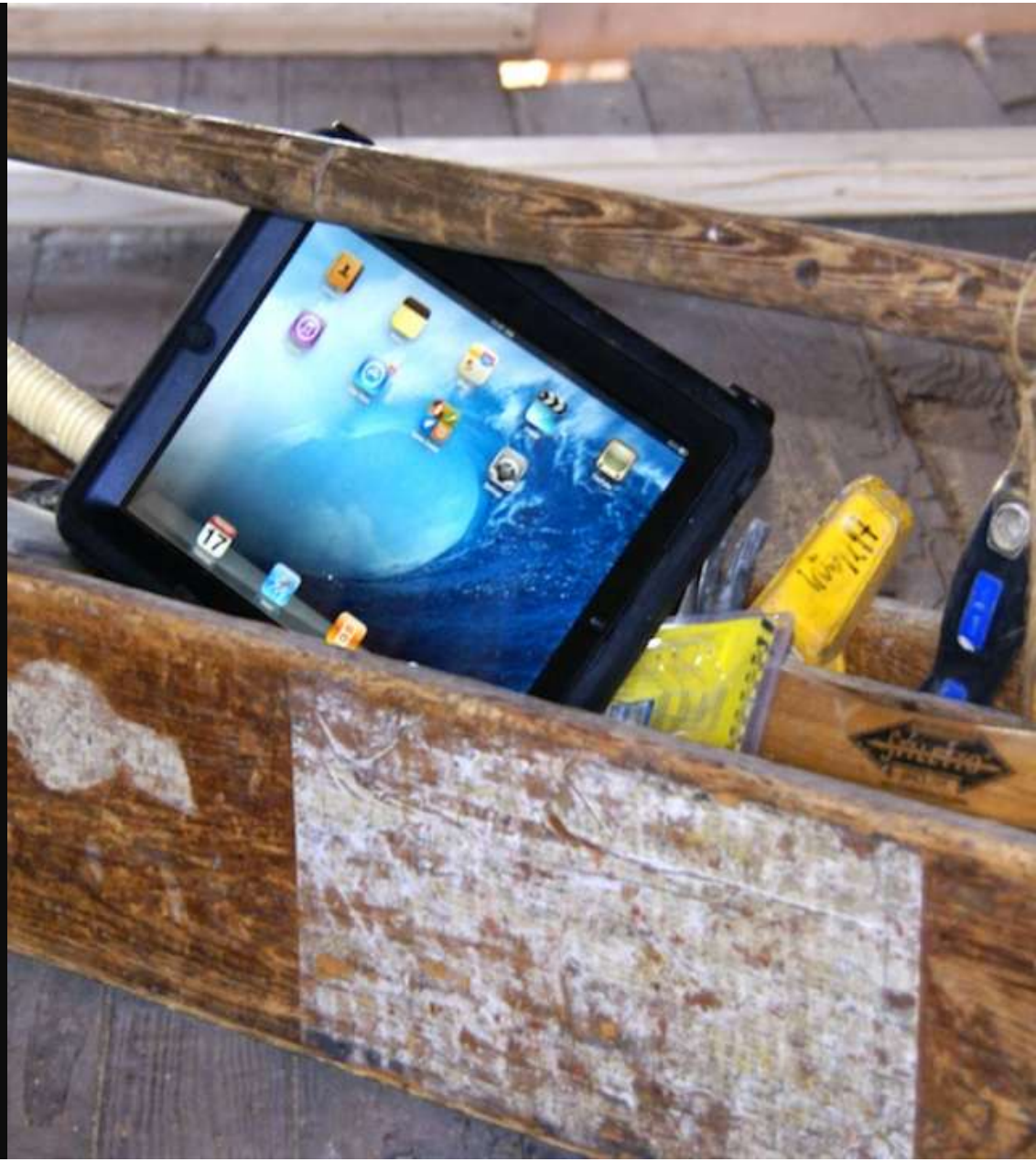


Employee Portal

Labor & Progress Tracking

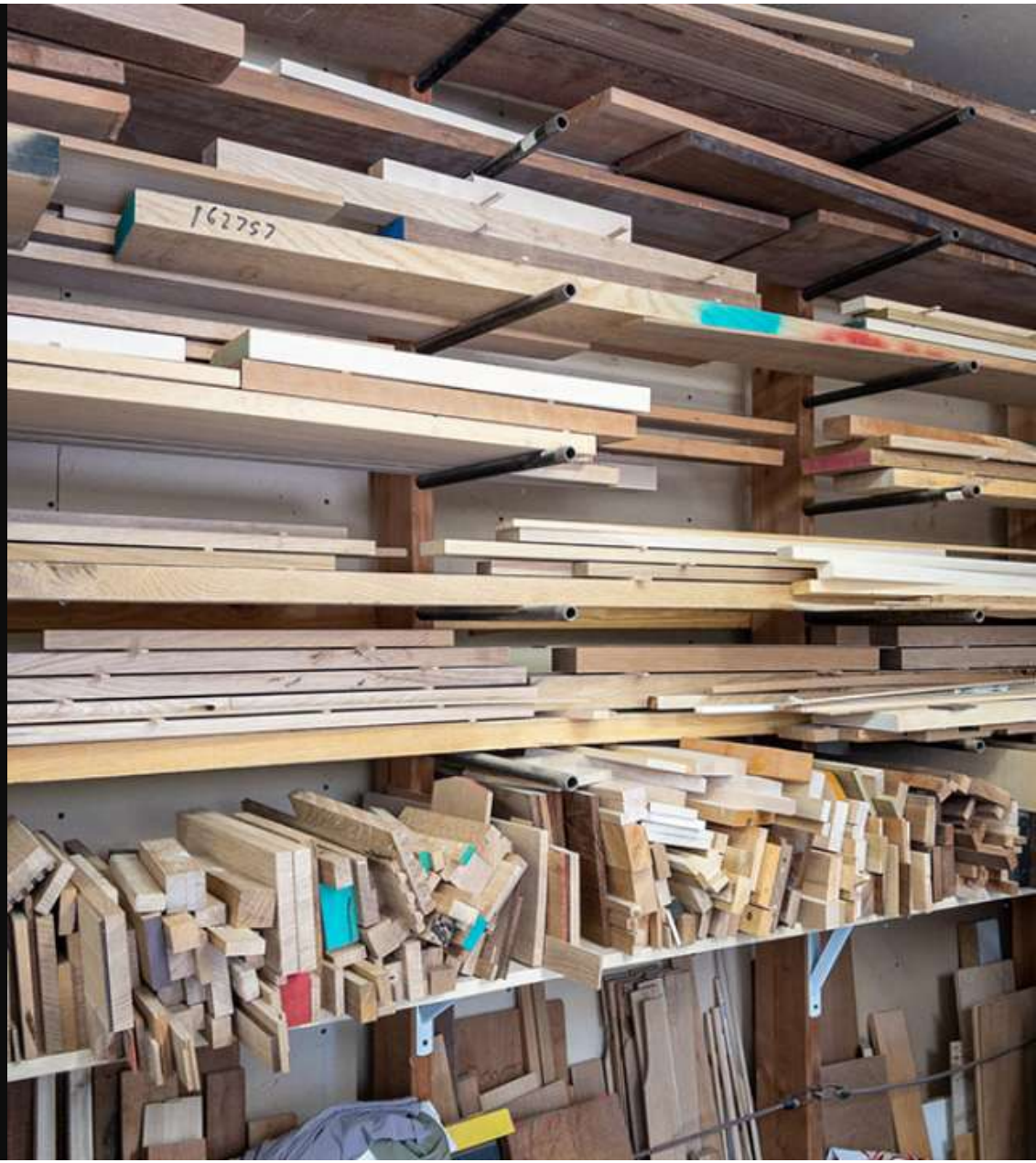
Artisans log in to see their specific assignments. This eliminates verbal instructions and ensures every task is documented.

Worklogs: Employees record their hours, which are then used by the system to auto-calculate wages, ensuring fairness and accuracy.



Inventory Intelligence

- **Material Catalog:** Tracking wood types, hardware, finishes, and consumables.
- **Auto-Stock Updates:** Automatic depletion of materials upon assignment to an order.
- **Low-Stock Awareness:** Real-time alerts when inventory falls below minimum thresholds.



User Role Framework



Admin

Full control over users, approvals, inventory, and comprehensive financial reports.



Customer

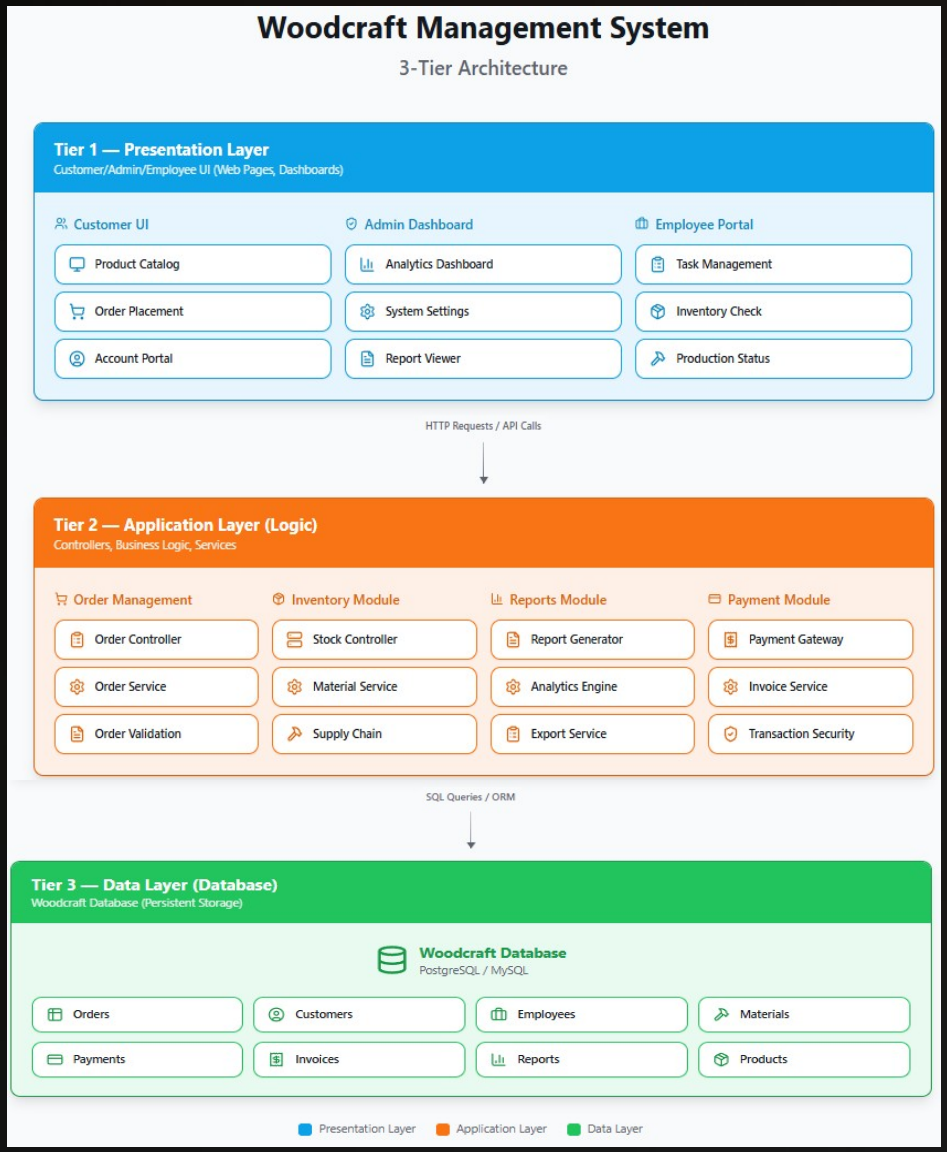
Places commissions, tracks order progress in real-time, and manages billing/receipts.



Employee

Views assigned tasks, logs daily work hours, and tracks wage calculations.

Architecture Diagram



Single Developer Framework



Technical Lead

Developing the core logic with React 18 and Node.js, ensuring a seamless full-stack integration.



Database Architect

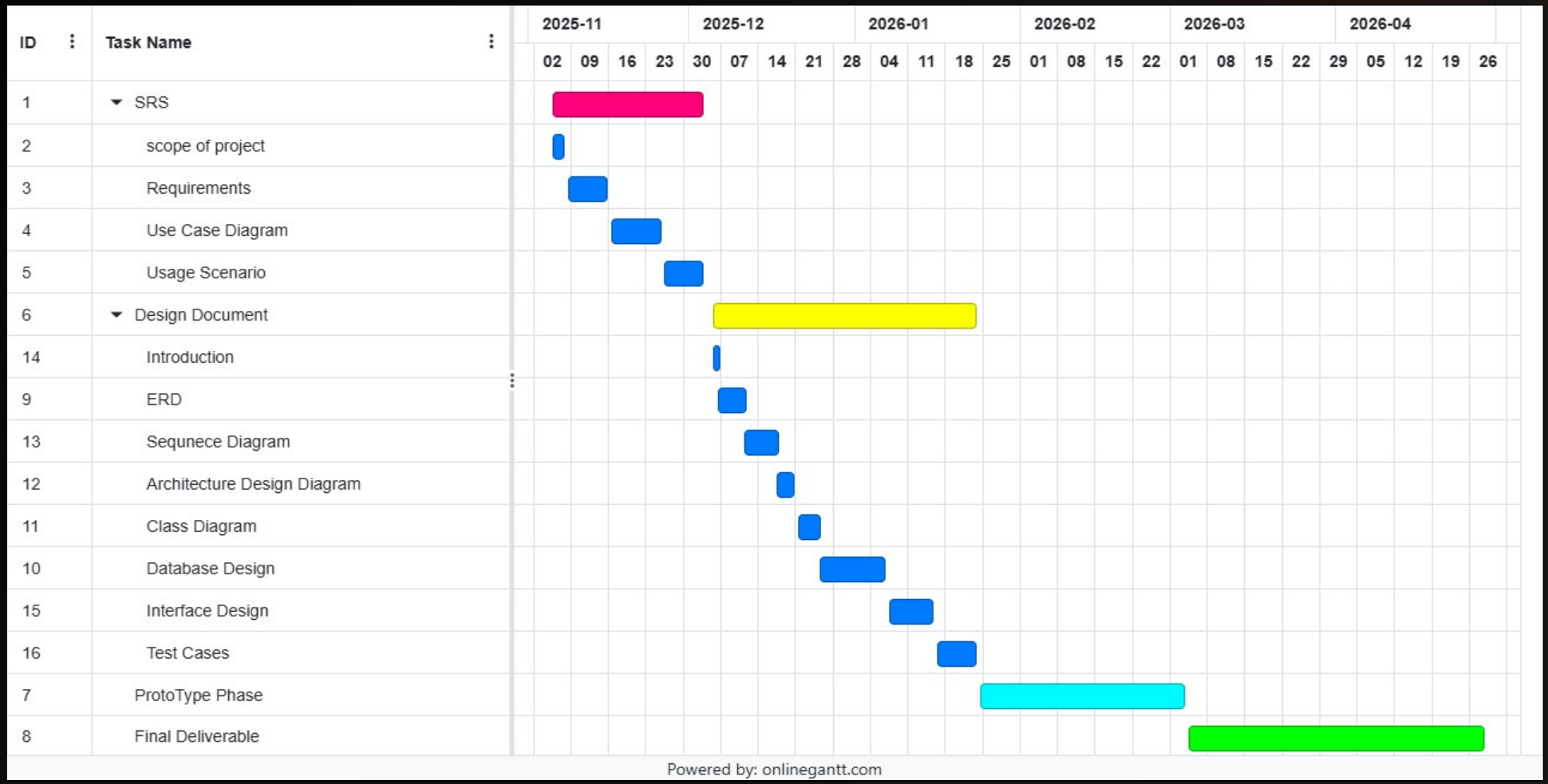
Designing MySQL relational structures for persistence, integrity, and complex billing queries.



UI/UX Designer

Crafting role-aware interfaces using Tailwind CSS and shadcn/ui for intuitive workshop management.

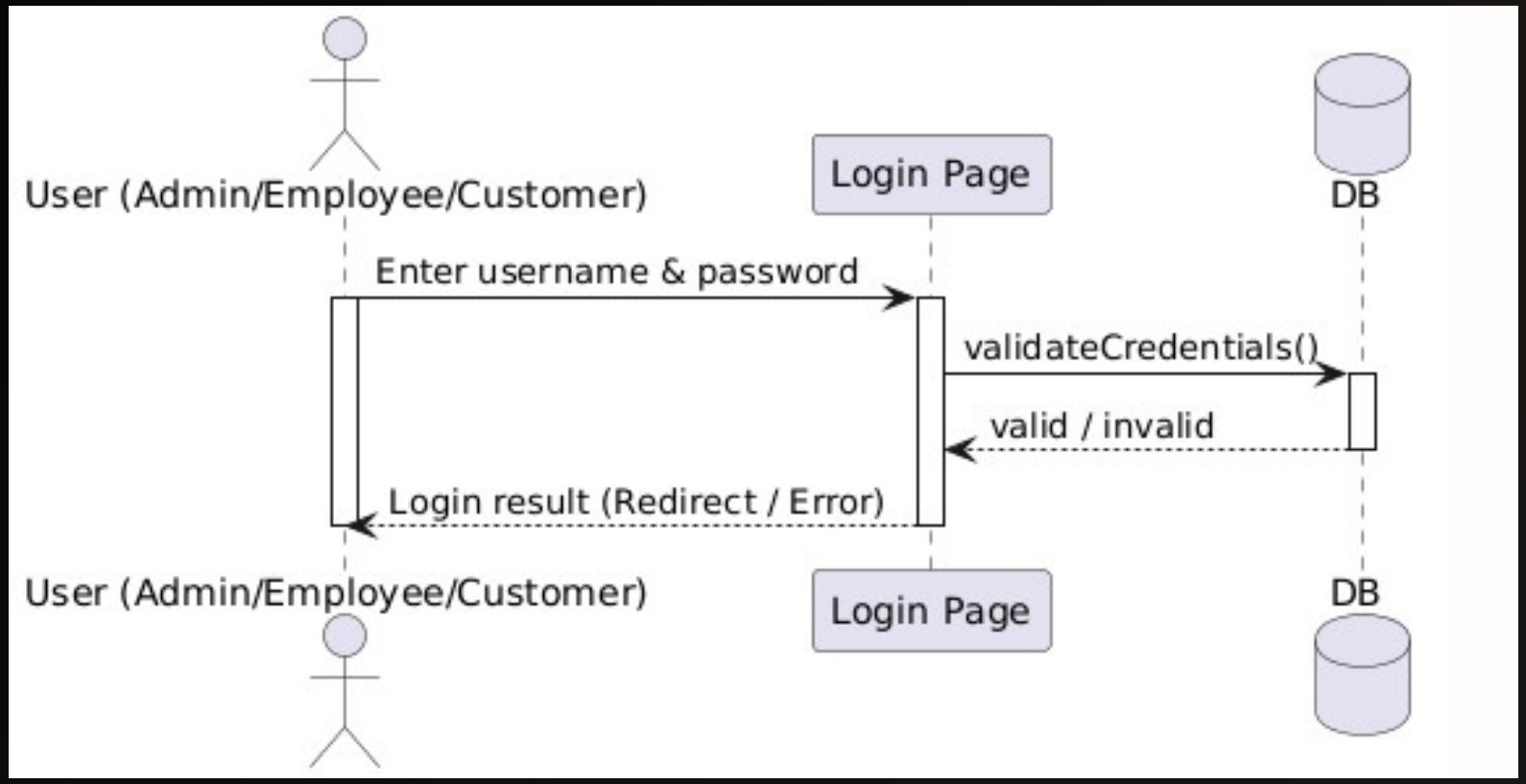
Project Schedule (Gantt Chart)



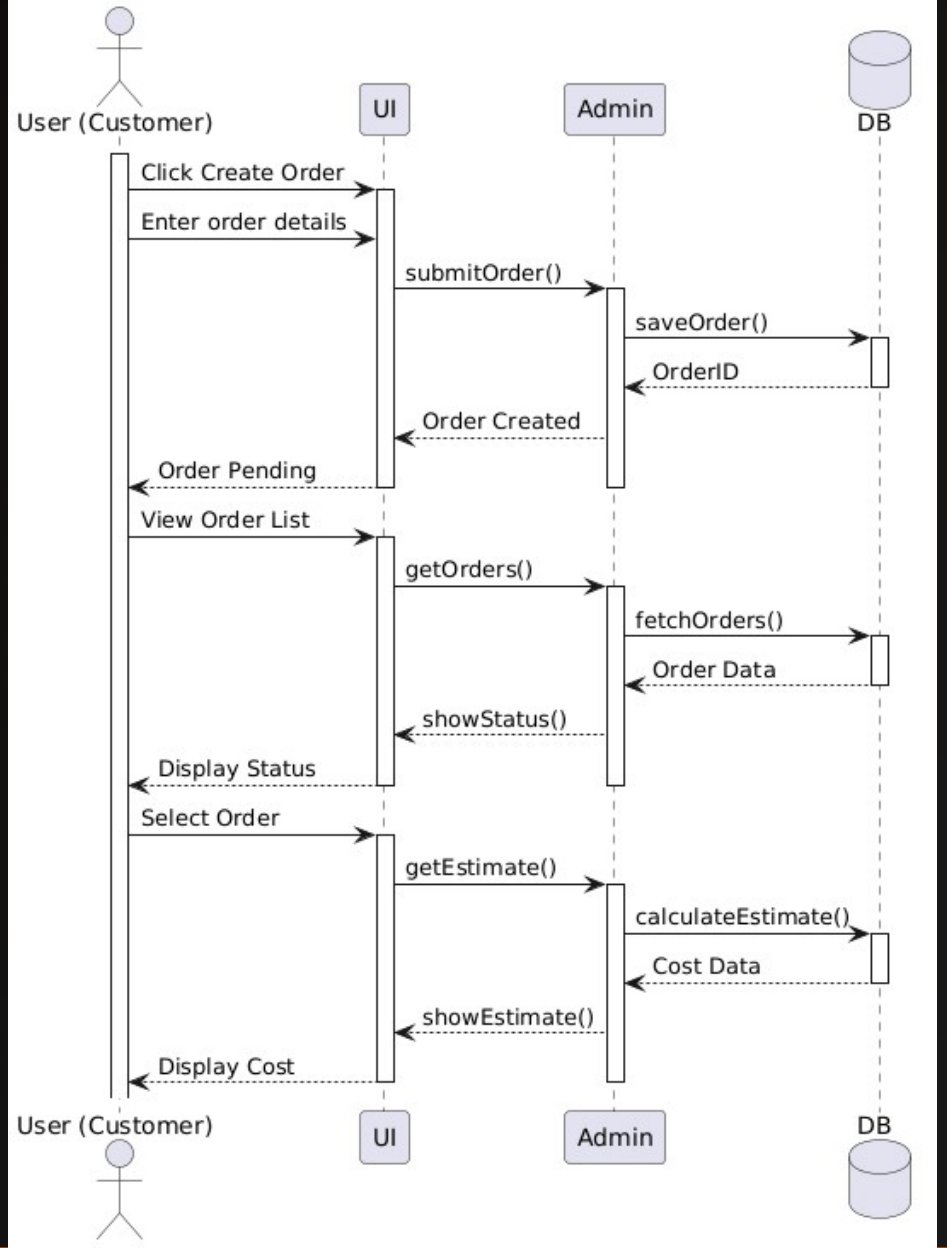
Sequence Diagrams of each Case

Sequence diagrams depict how system components interact over time to complete a particular use case.

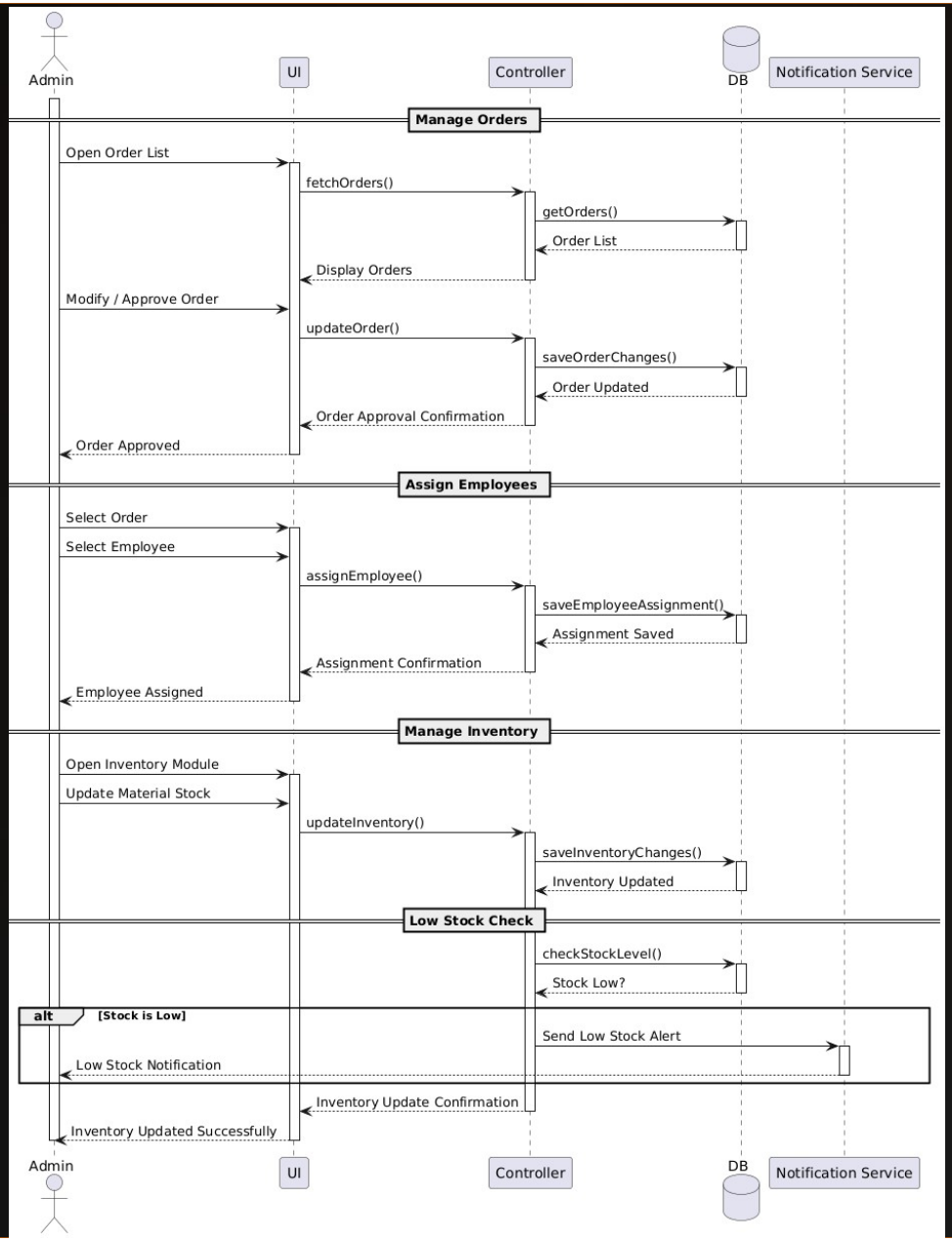
Sequence diagrams (Login/Register)



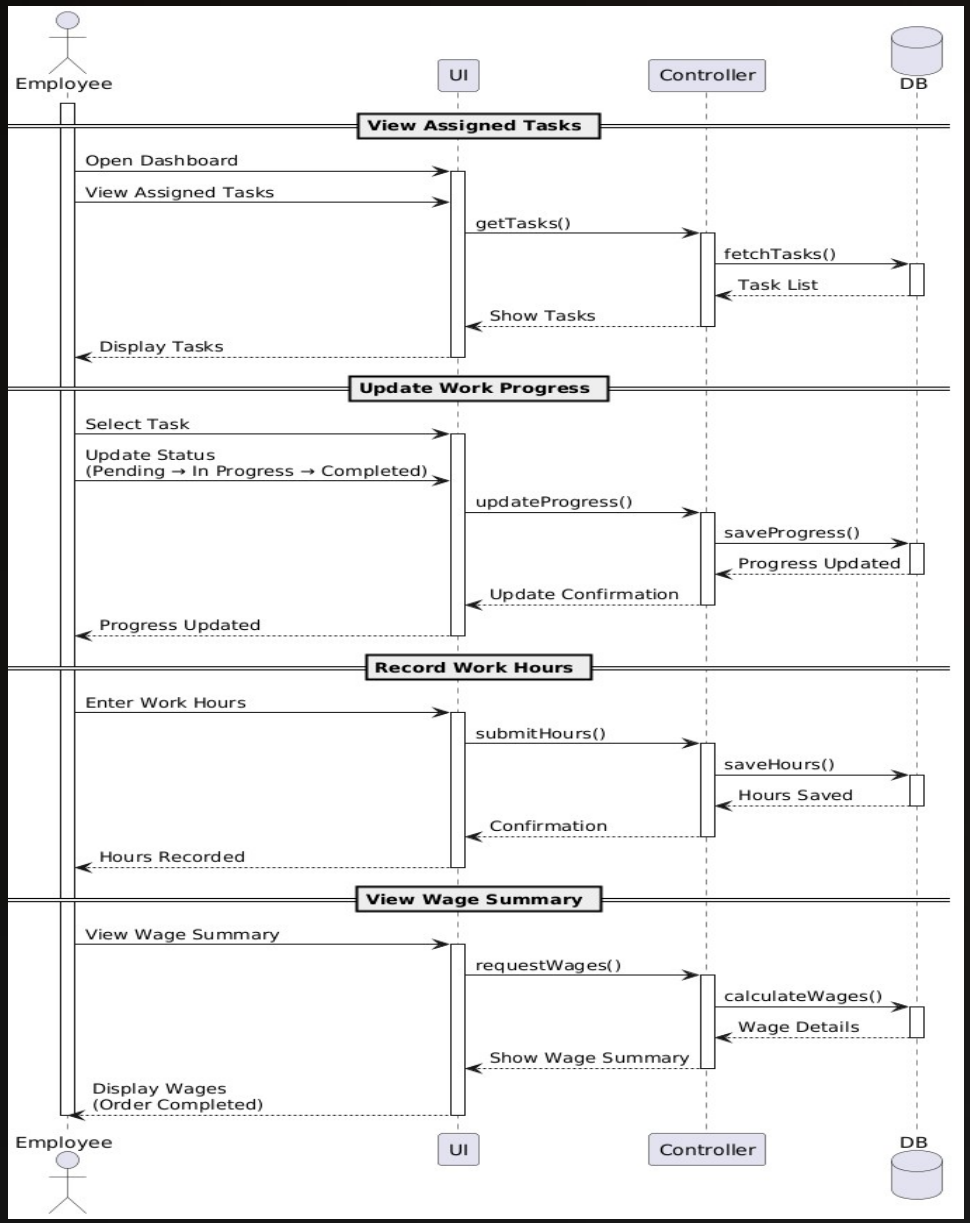
Sequence diagrams
(Customer Initiates order)



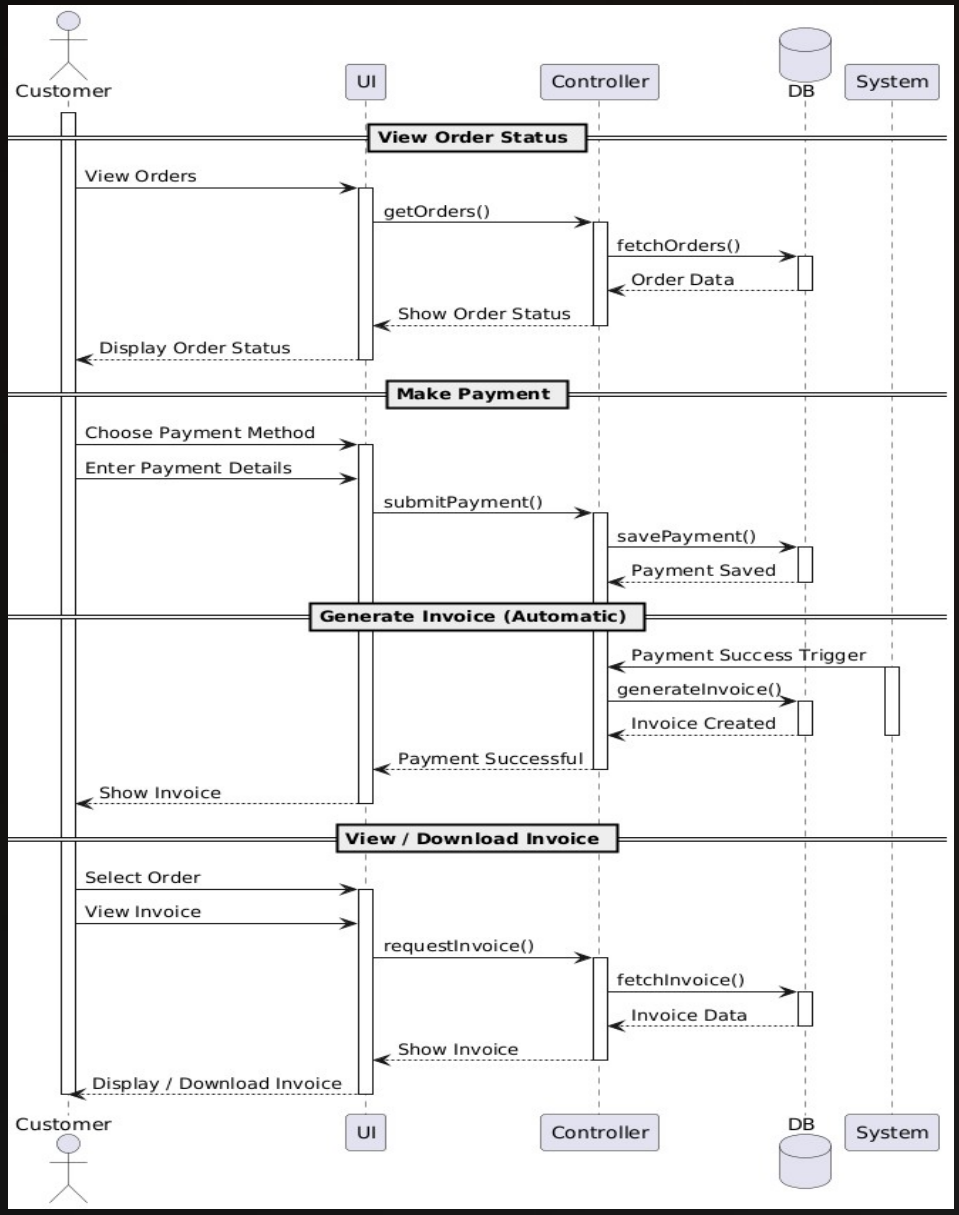
Sequence diagrams
(Admin Order Processing)



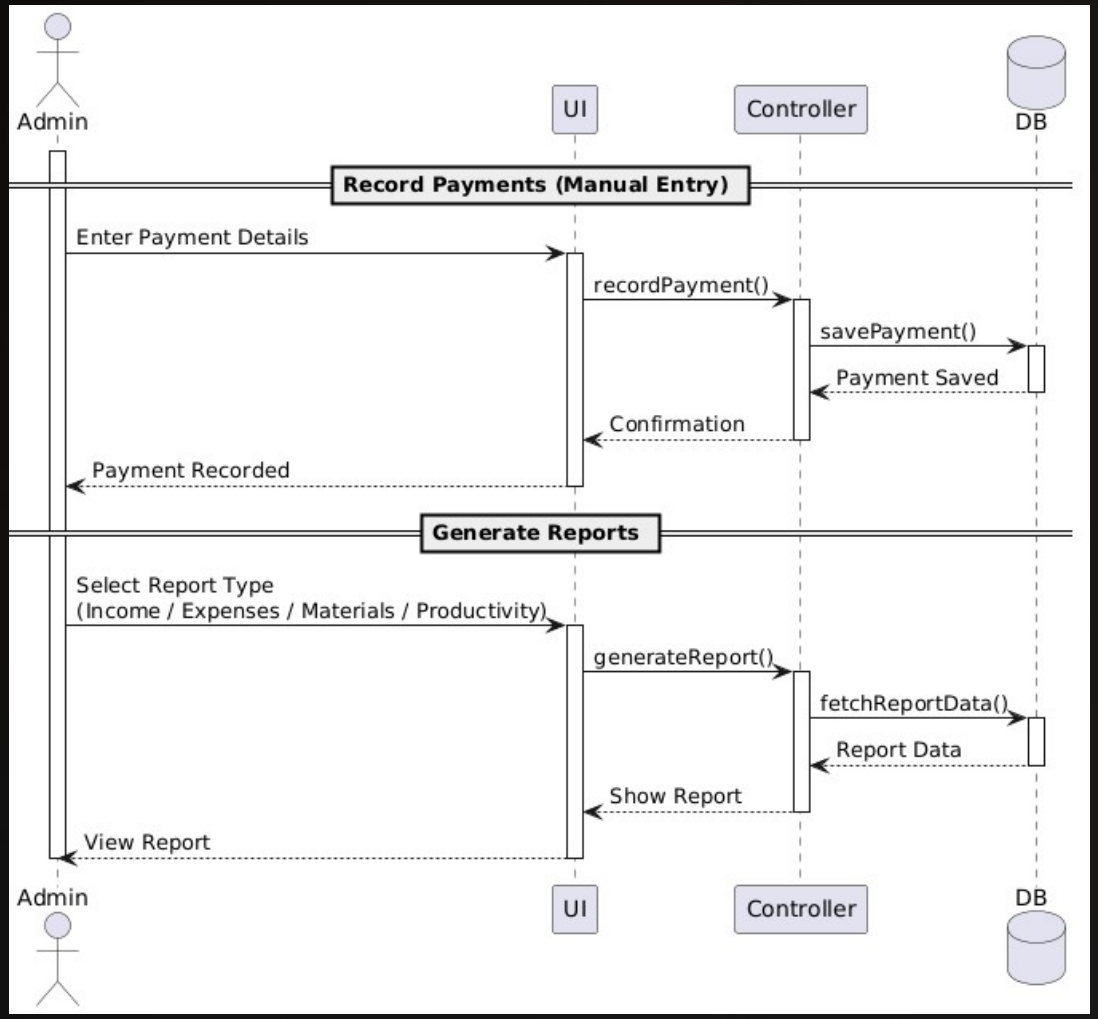
Sequence diagrams
(Employee's Order)



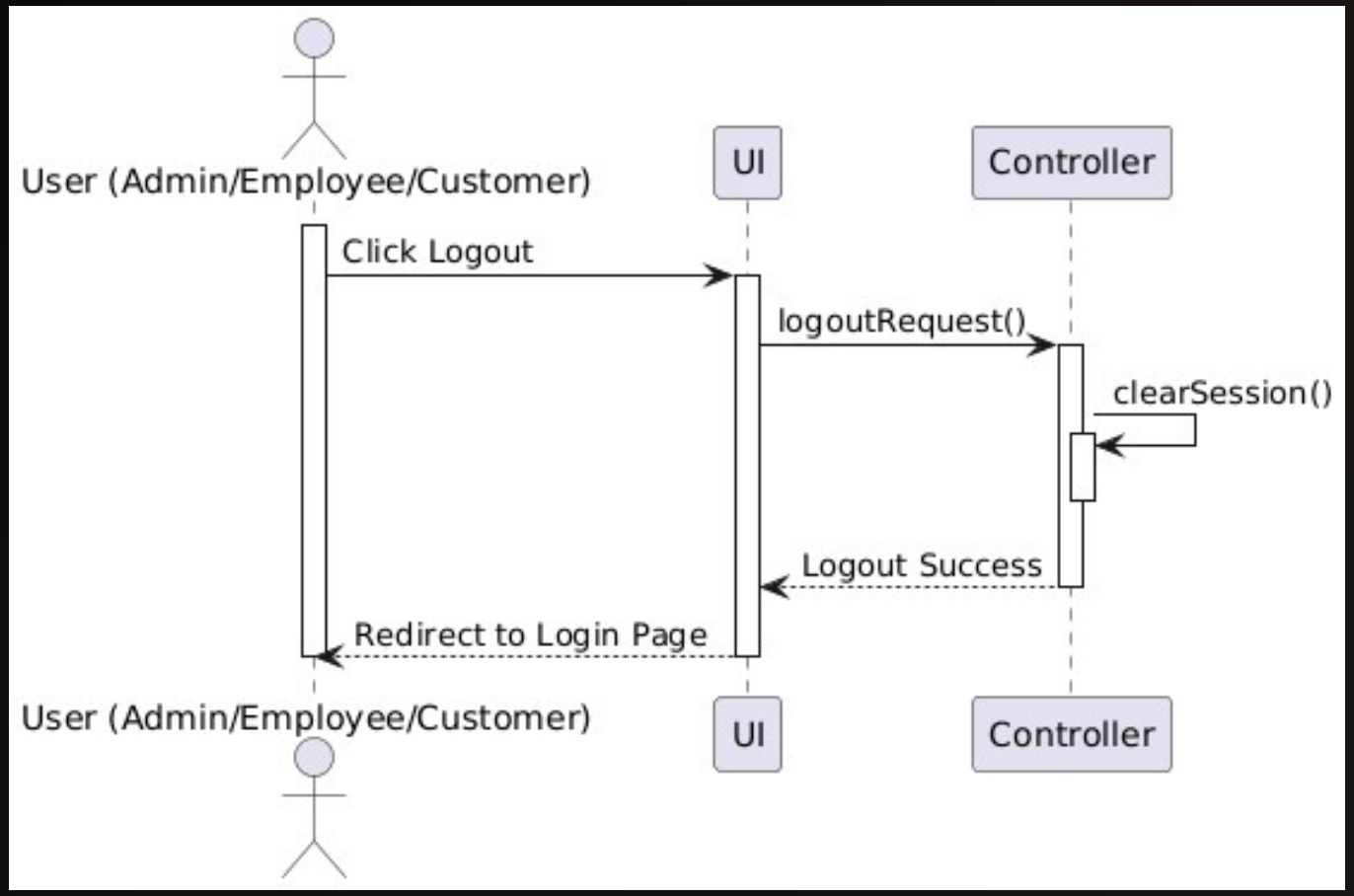
Sequence diagrams
(Payment by Customer)



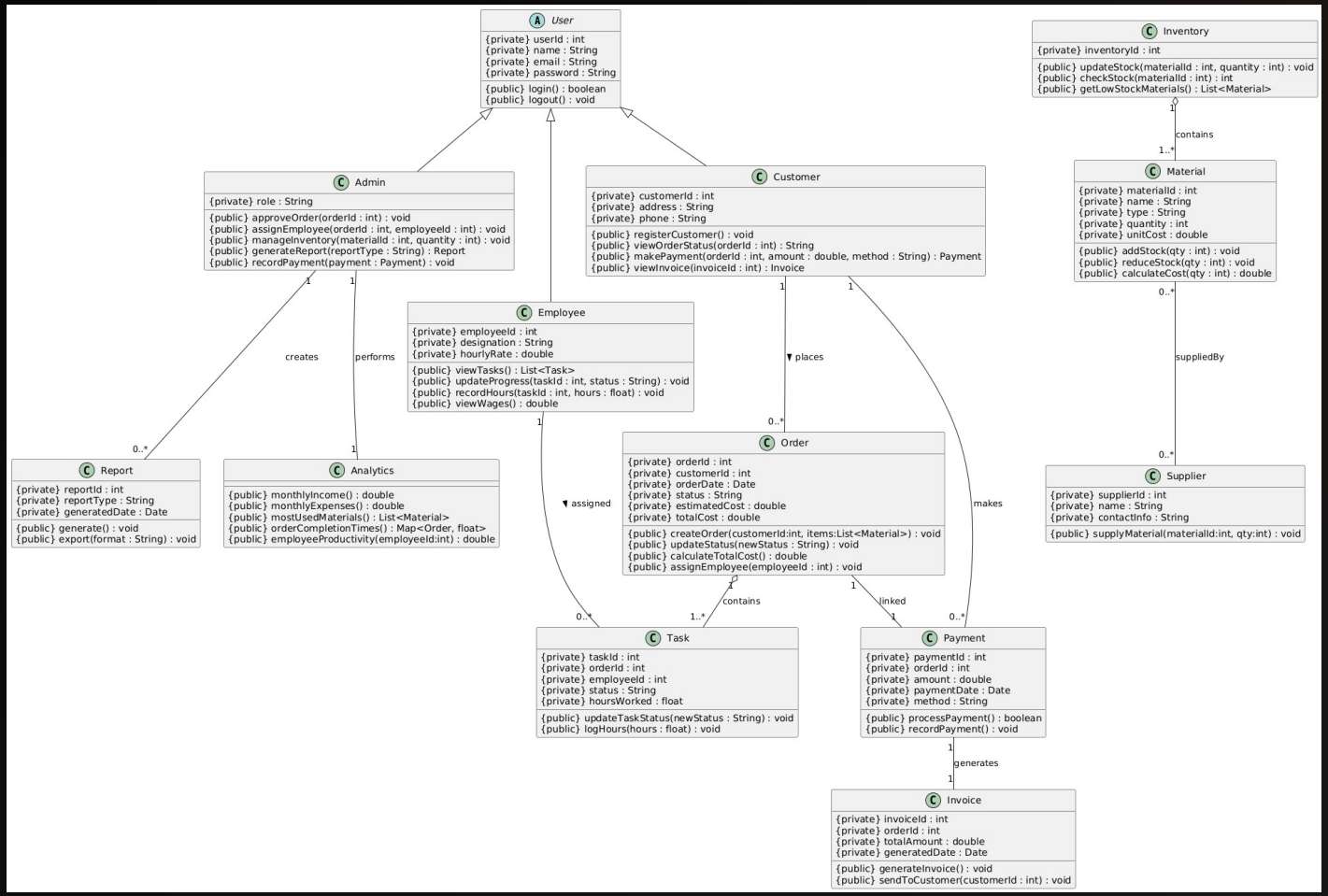
Sequence diagrams
(Reports generation after payment)



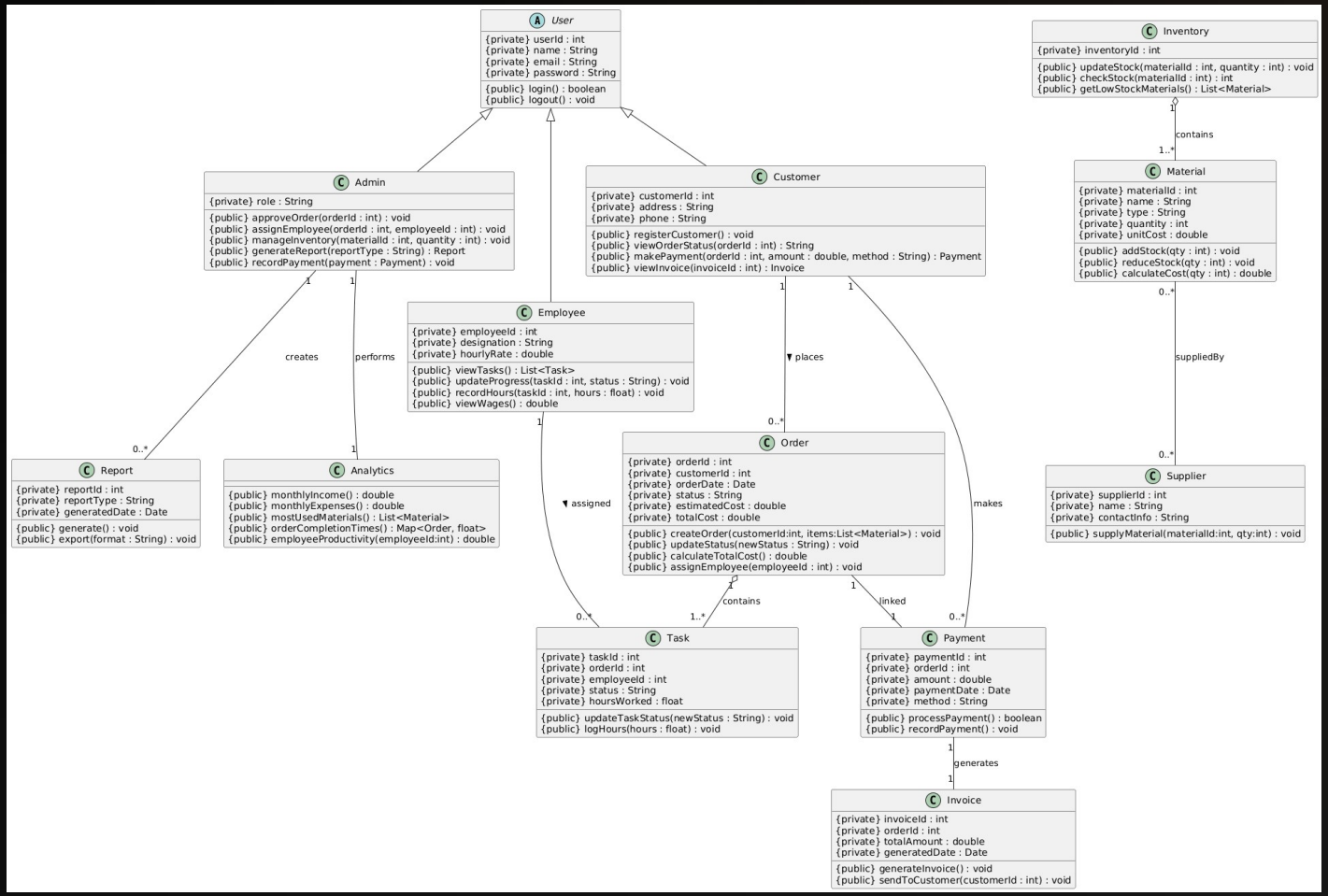
Sequence diagrams
(Logout)



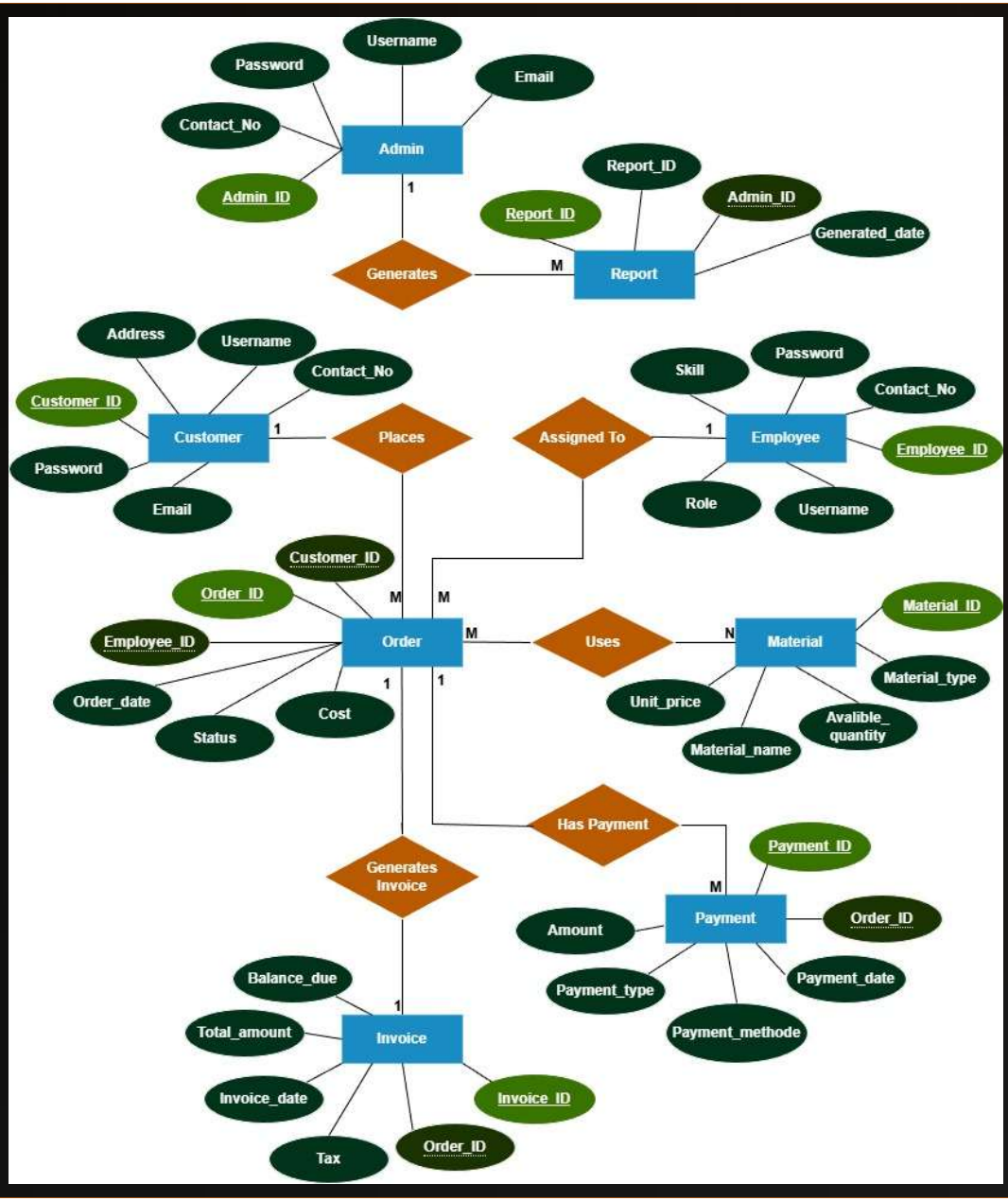
Class Diagram



Class Diagram



Entity-Relationship Diagram



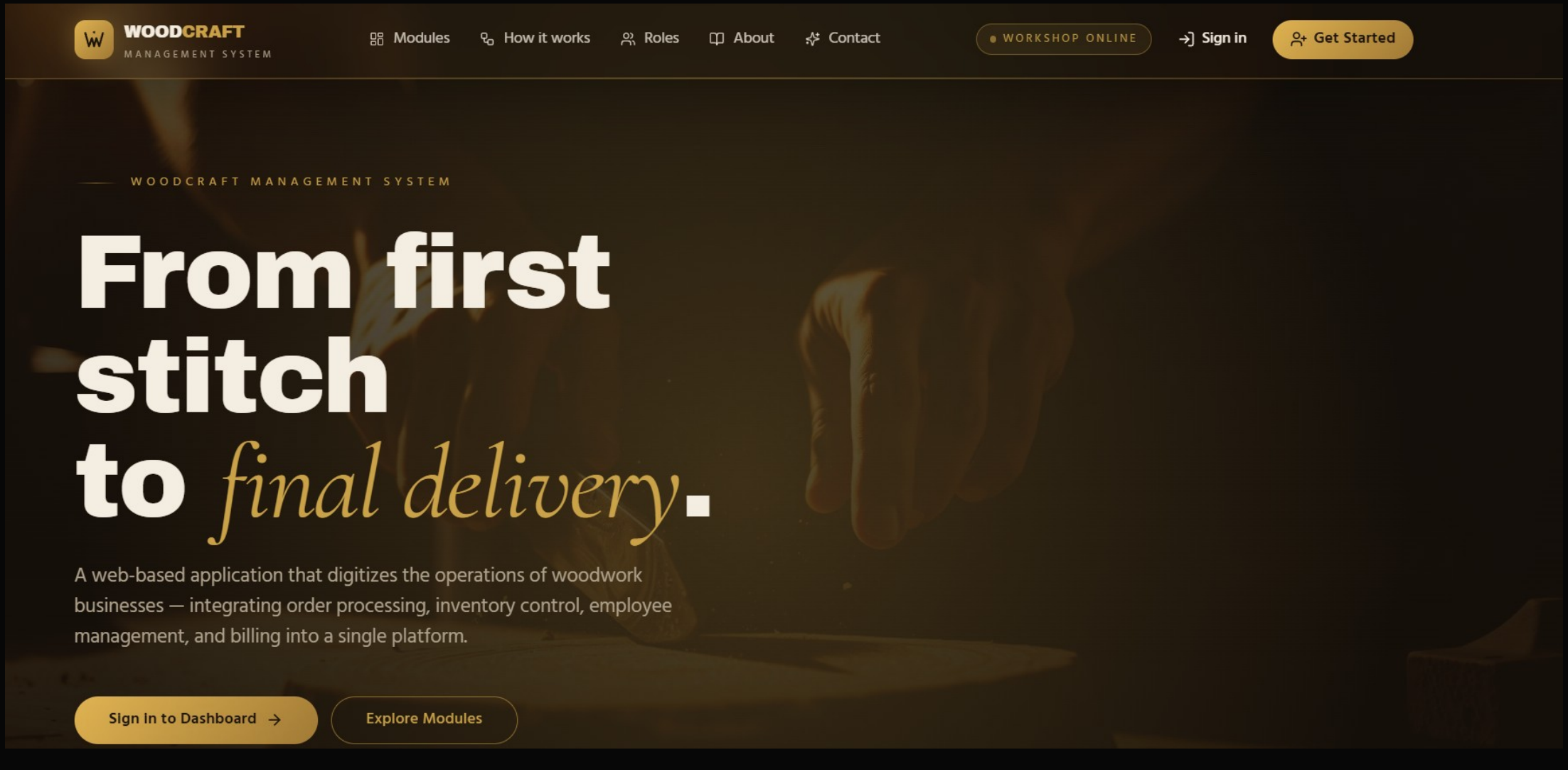
Database Diagram



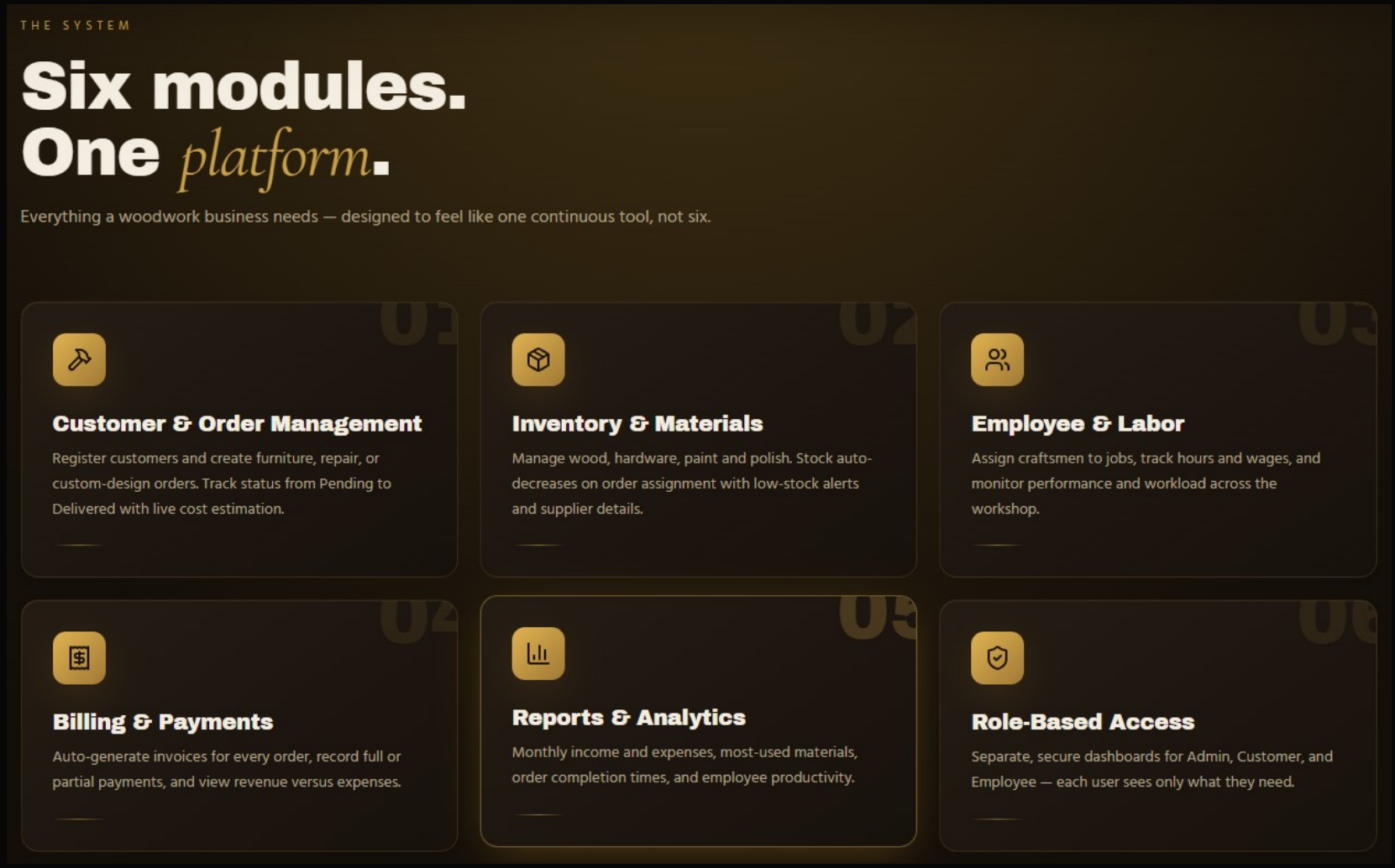
Graphical User Interface

Woodcraft Web Application

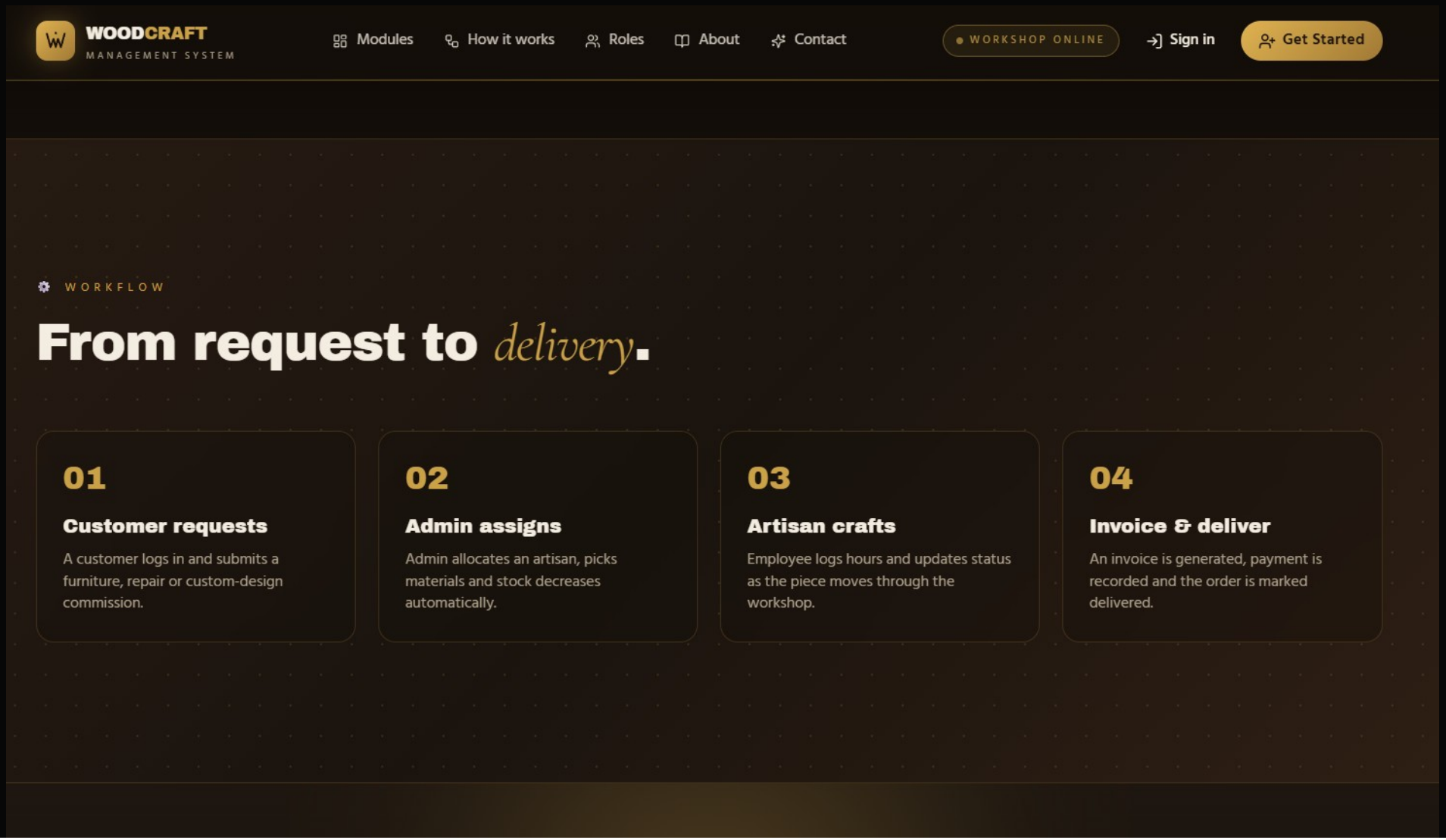
HomePage:


WOODCRAFT
MANAGEMENT SYSTEM
Modules
How it works
Roles
About
Contact
WORKSHOP ONLINE
Sign in
Get Started
WOODCRAFT MANAGEMENT SYSTEM
From first stitch to final delivery.
A web-based application that digitizes the operations of woodwork businesses — integrating order processing, inventory control, employee management, and billing into a single platform.
Sign In to Dashboard
Explore Modules

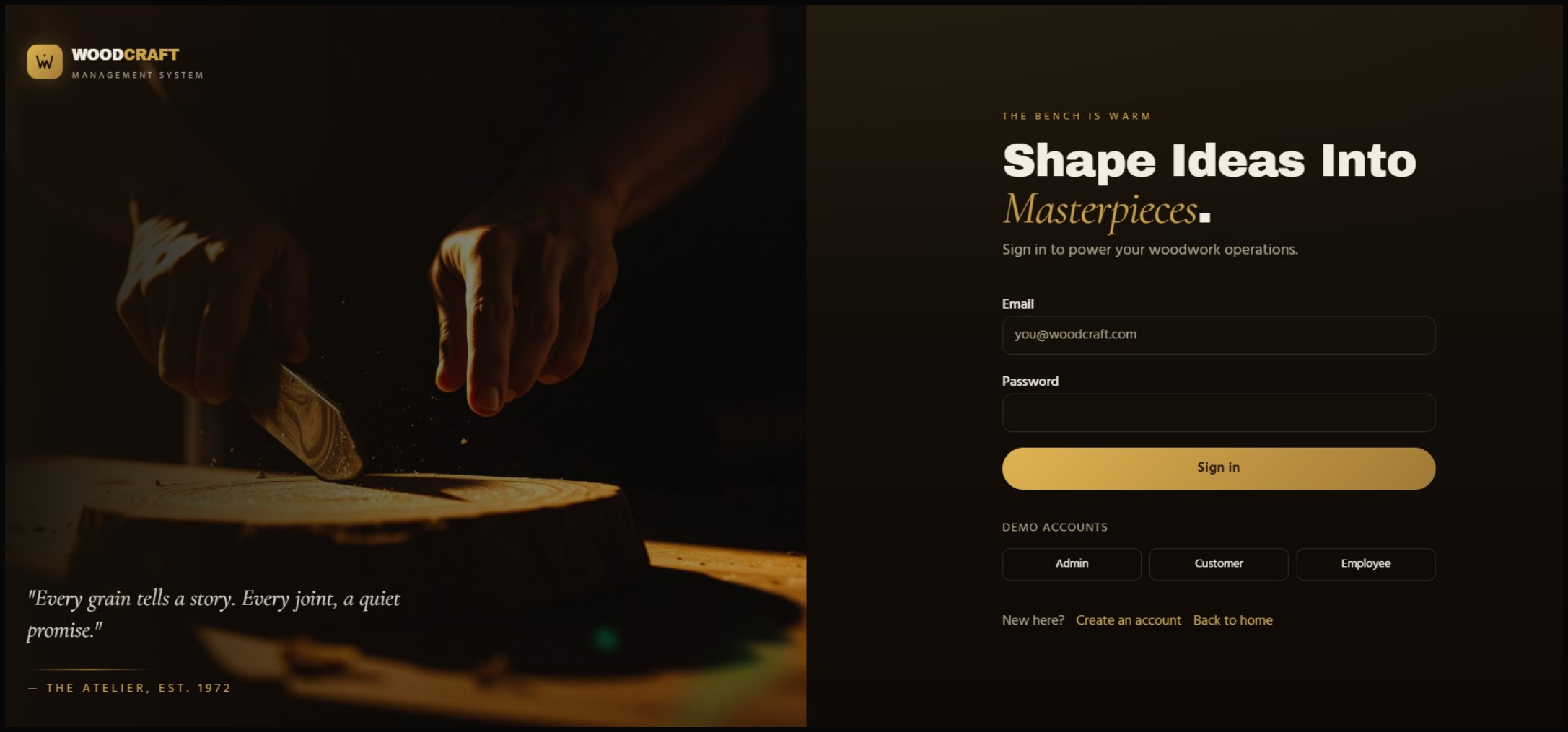
HomePage:


THE SYSTEM
Six modules.
One platform.
Everything a woodwork business needs — designed to feel like one continuous tool, not six.
Customer & Order Management
Register customers and create furniture, repair, or custom-design orders. Track status from Pending to Delivered with live cost estimation.
Inventory & Materials
Manage wood, hardware, paint and polish. Stock auto-decreases on order assignment with low-stock alerts and supplier details.
Employee & Labor
Assign craftsmen to jobs, track hours and wages, and monitor performance and workload across the workshop.
Billing & Payments
Auto-generate invoices for every order, record full or partial payments, and view revenue versus expenses.
Reports & Analytics
Monthly income and expenses, most-used materials, order completion times, and employee productivity.
Role-Based Access
Separate, secure dashboards for Admin, Customer, and Employee — each user sees only what they need.

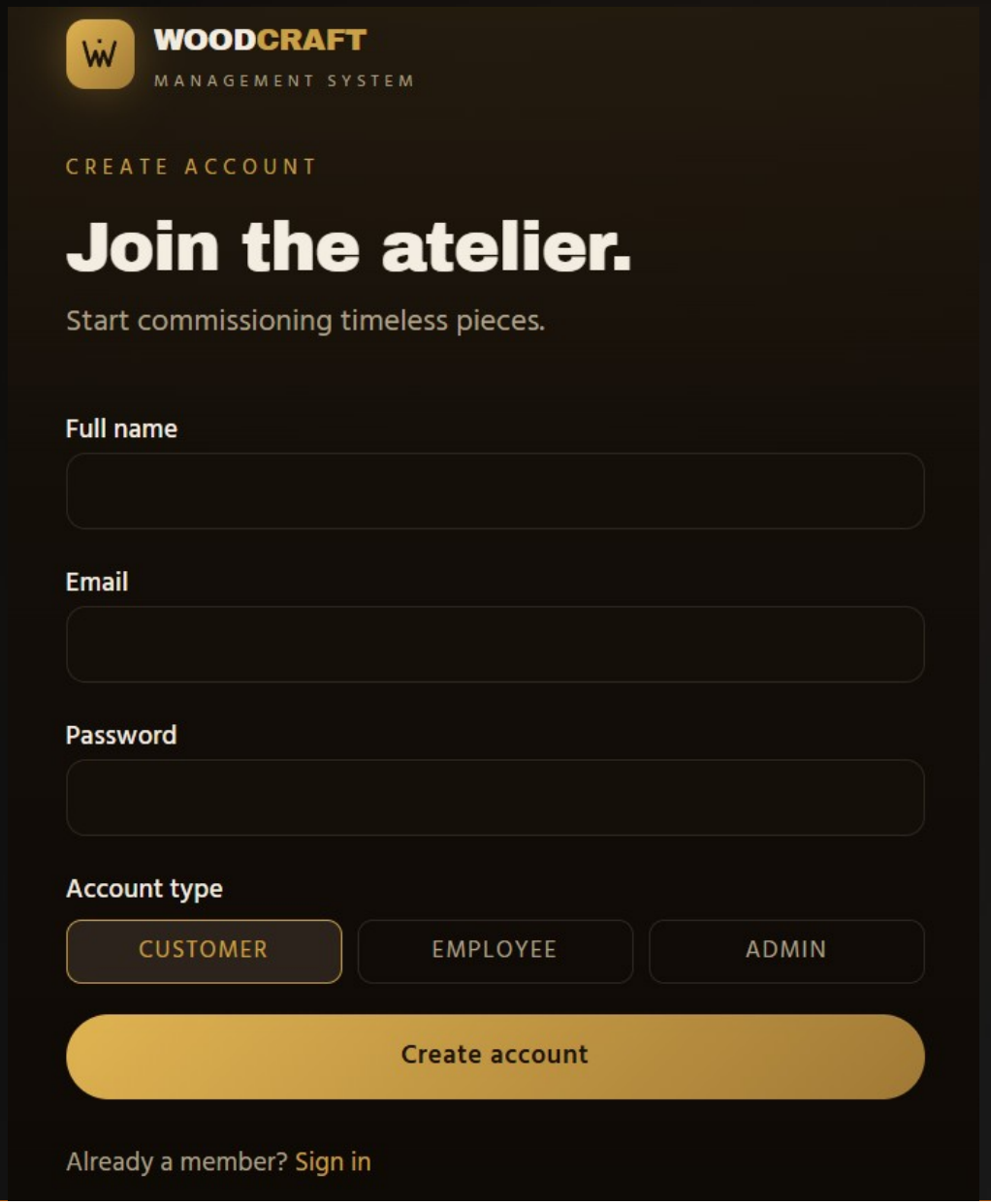
HomePage:


WOODCRAFT
MANAGEMENT SYSTEM
Modules
How it works
Roles
About
Contact
WORKSHOP ONLINE
Sign in
Get Started
WORKFLOW
From request to delivery.
01
Customer requests
A customer logs in and submits a furniture, repair or custom-design commission.
02
Admin assigns
Admin allocates an artisan, picks materials and stock decreases automatically.
03
Artisan crafts
Employee logs hours and updates status as the piece moves through the workshop.
04
Invoice & deliver
An invoice is generated, payment is recorded and the order is marked delivered.

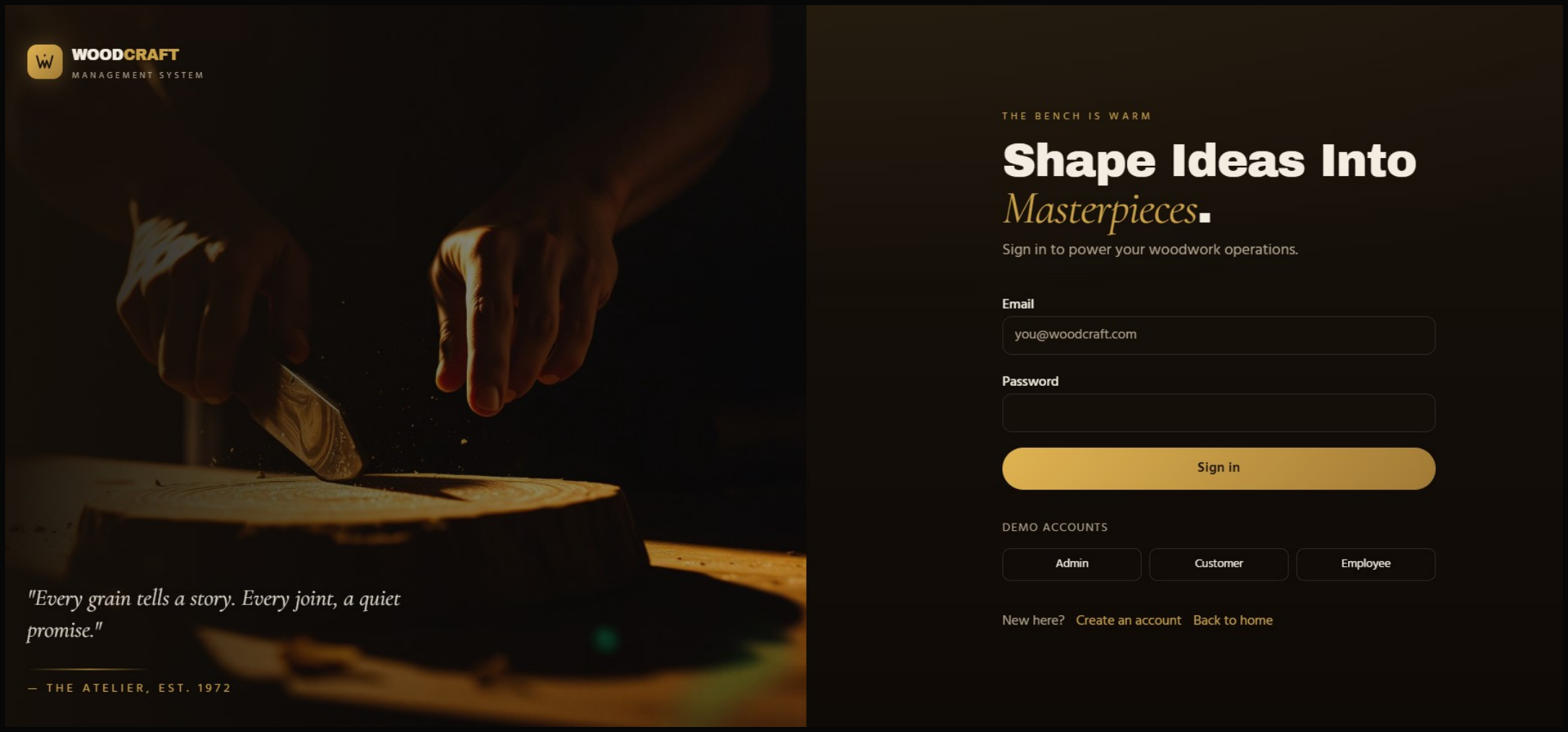
Login Page:


WOODCRAFT
MANAGEMENT SYSTEM
"Every grain tells a story. Every joint, a quiet promise."
— THE ATELIER, EST. 1972
THE BENCH IS WARM
Shape Ideas Into
Masterpieces.
Sign in to power your woodwork operations.
Email
you@woodcraft.com
Password
Sign in
DEMO ACCOUNTS
Admin
Customer
Employee
New here? Create an account Back to home

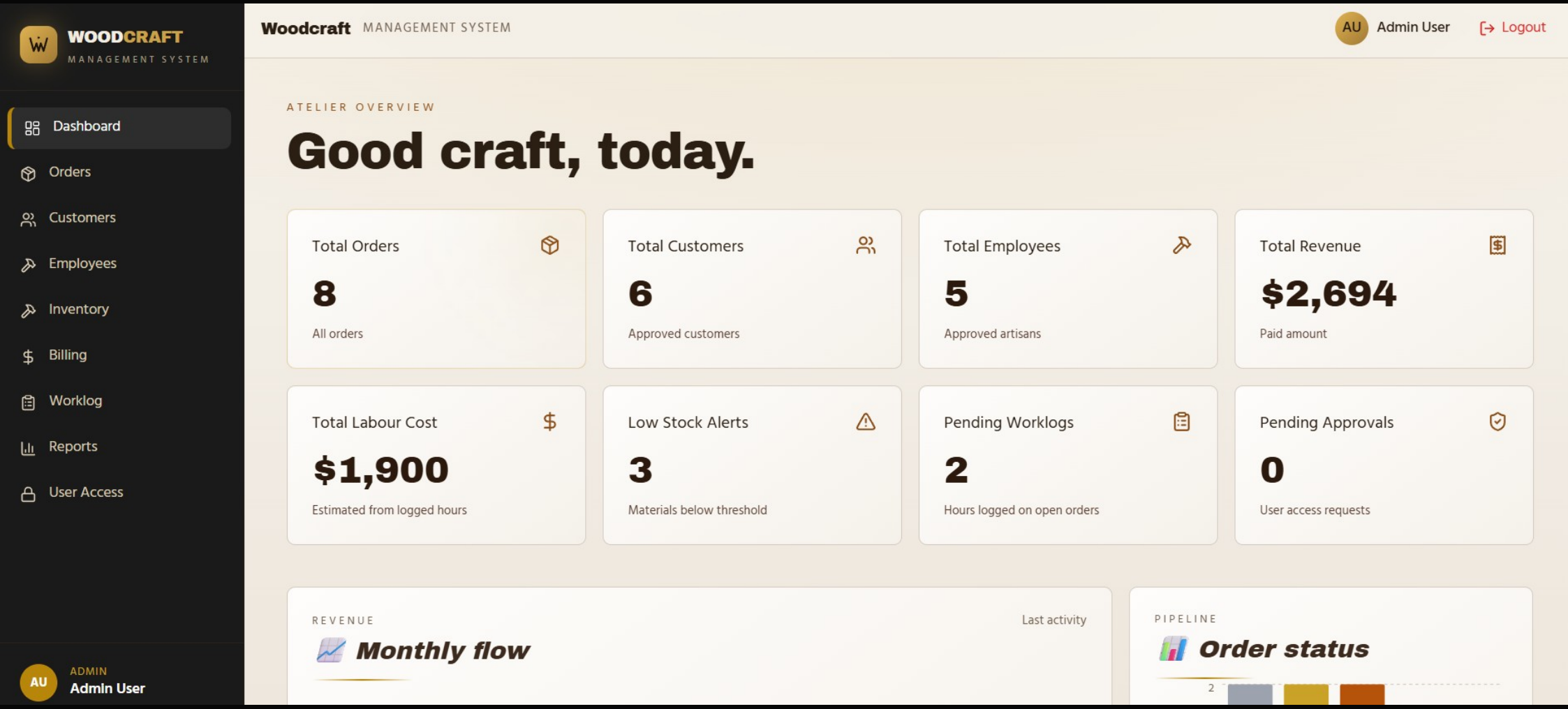
Register New Accounts



Login Page:


WOODCRAFT
MANAGEMENT SYSTEM
"Every grain tells a story. Every joint, a quiet promise."
— THE ATELIER, EST. 1972
THE BENCH IS WARM
Shape Ideas Into
Masterpieces.
Sign in to power your woodwork operations.
Email
you@woodcraft.com
Password
Sign in
DEMO ACCOUNTS
Admin
Customer
Employee
New here? Create an account Back to home

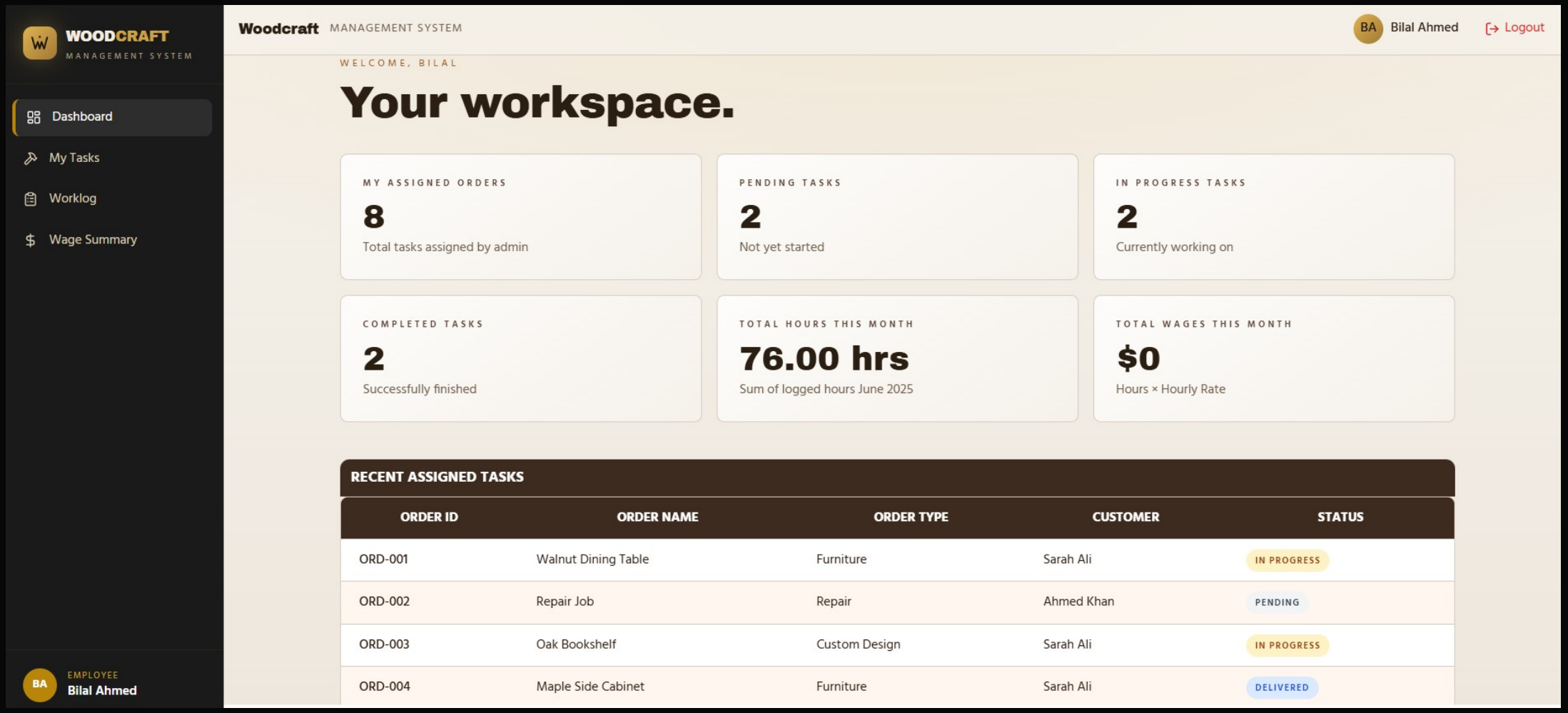
Admin Account



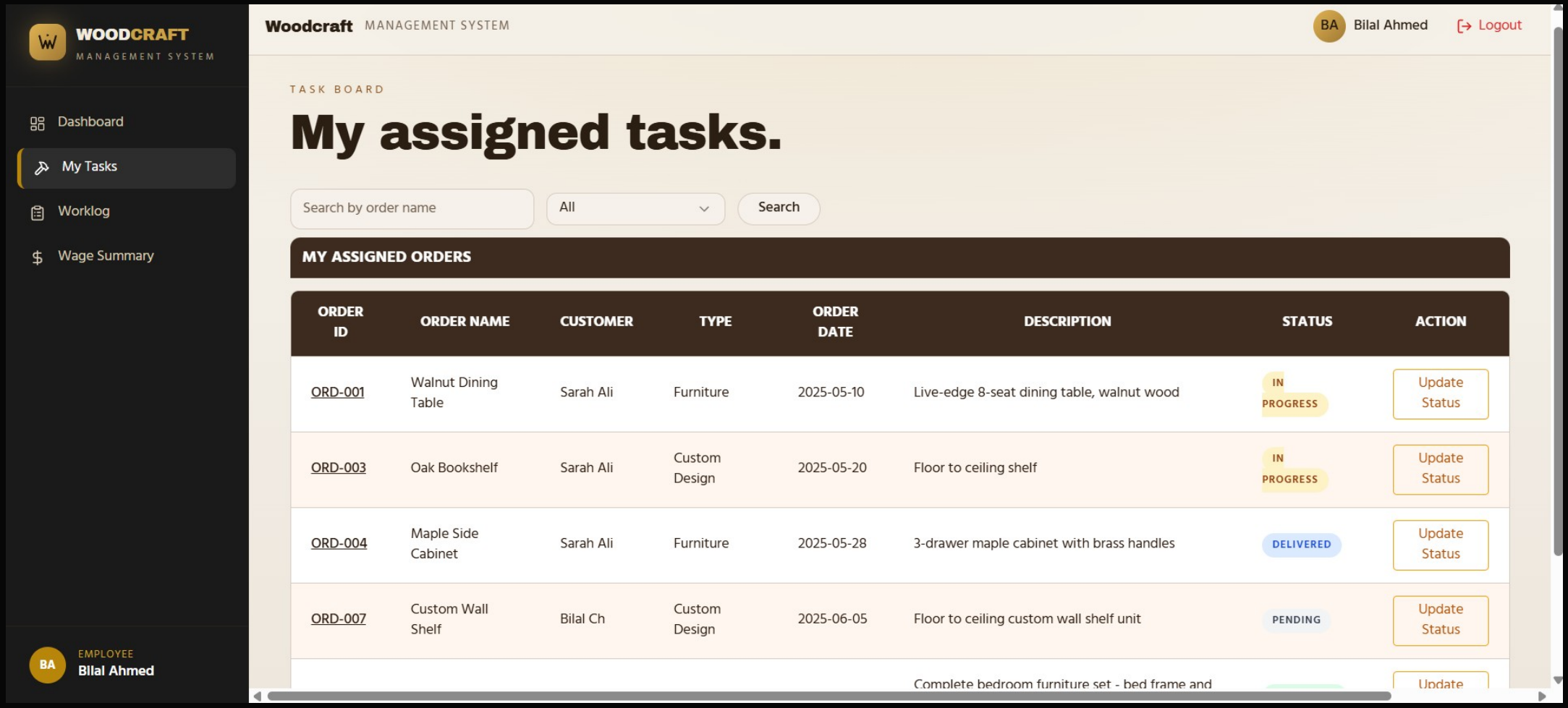
Admin Account



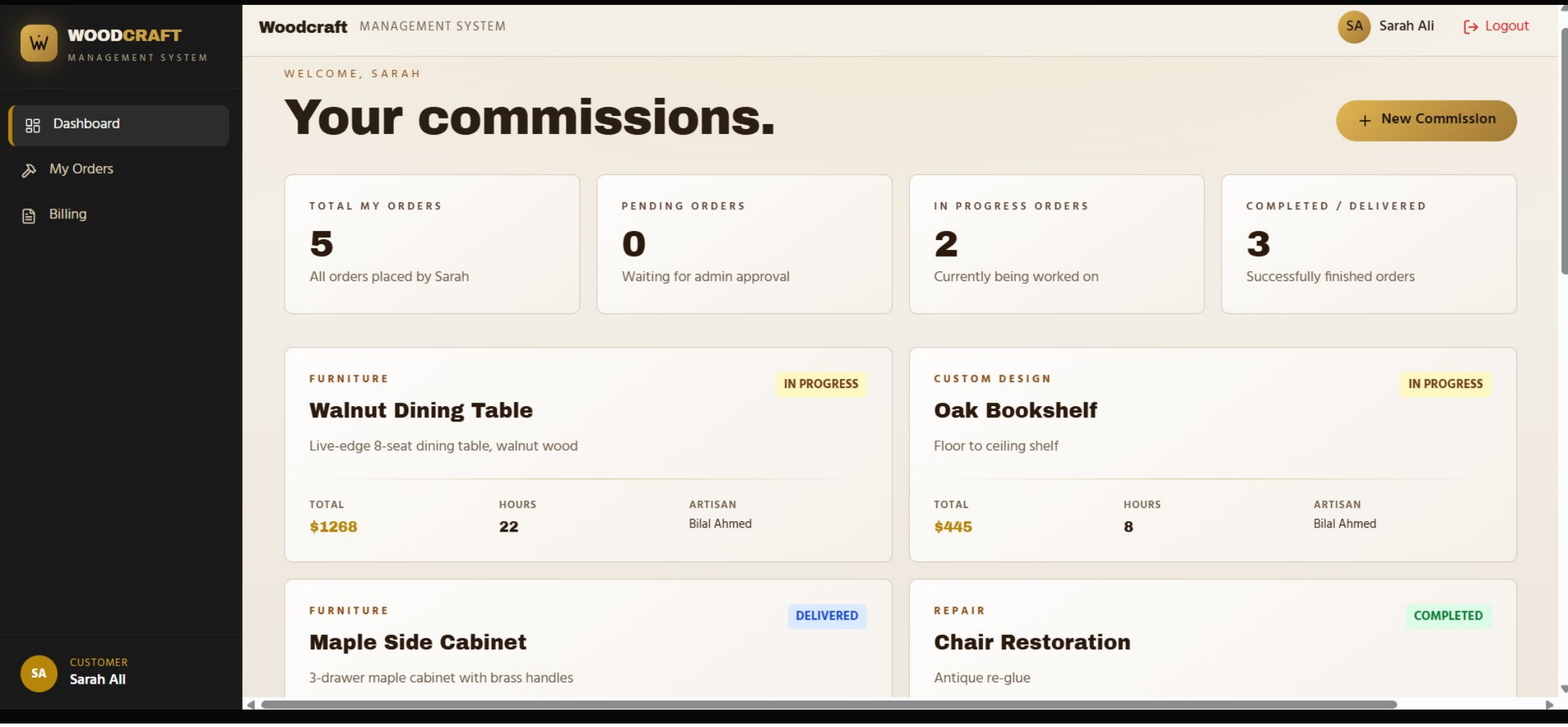
Employee Account



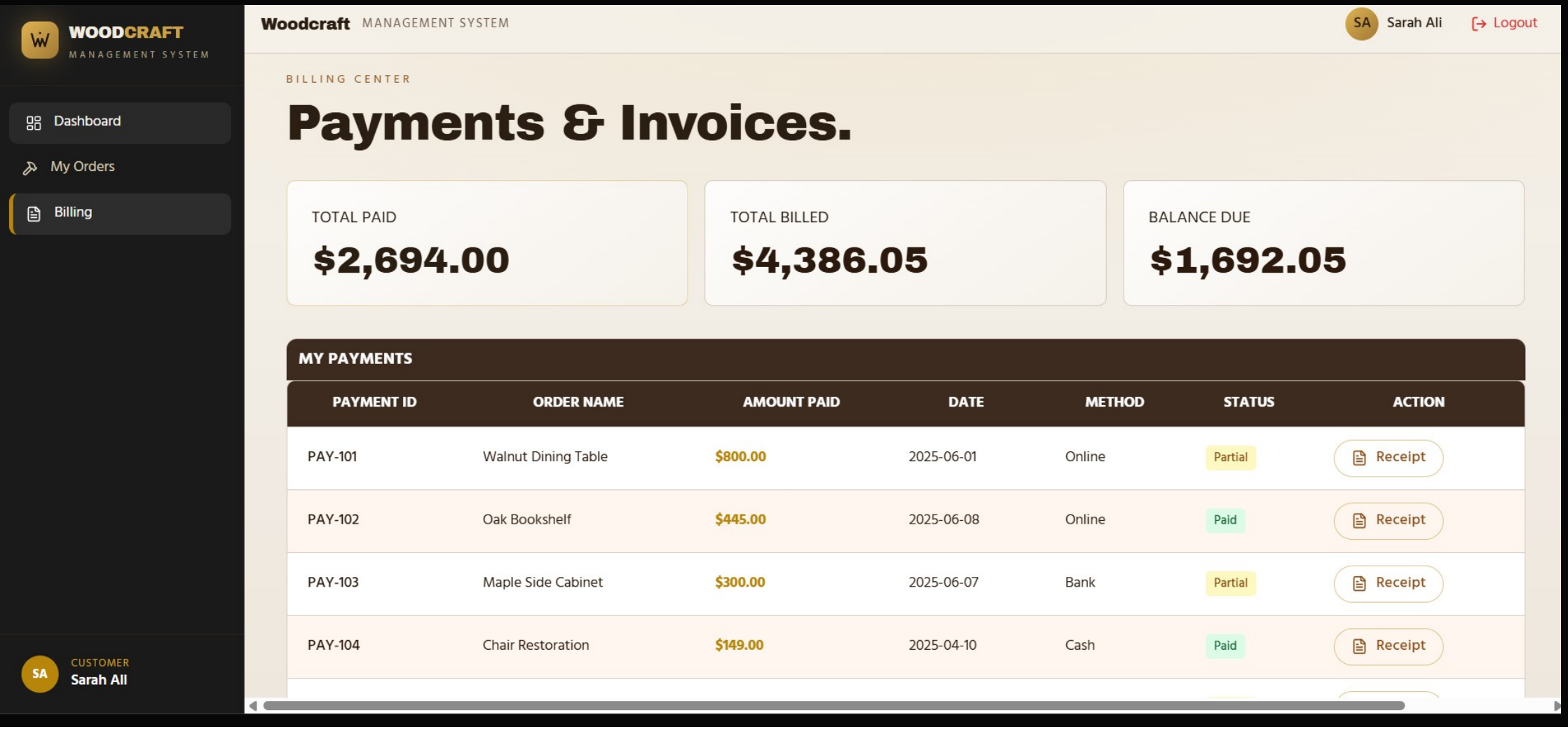
Employee Account



Customer Account



Customer Account

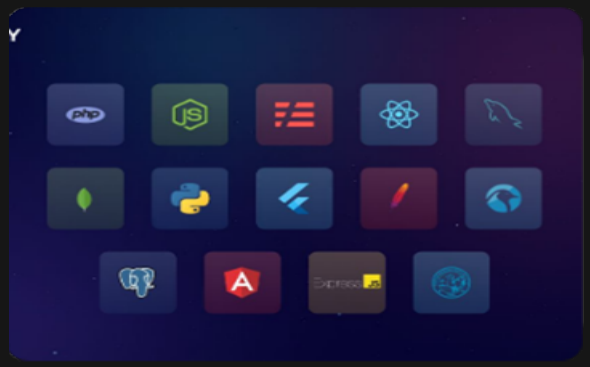


Tool & Modern Tech Stack

- **Frontend:** React 18, TypeScript, Vite, and Tailwind CSS for a performant UI.
- **Backend:** Node.js & Express.js reference implementation.
- **Storage:** MySQL with Relational Schema for data integrity.
- **Security:** JWT for authentication and BCrypt for password hashing.



The Tech Stack



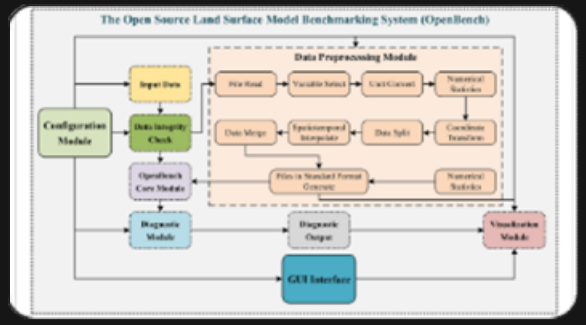
Frontend

React 18, Vite, TypeScript, Tailwind CSS, shadcn/ui



Backend

Node.js, Express, JWT Auth, Bcrypt Security



Database

MySQL Relational Schema for persistent storage


Thank You
VU
Virtual University